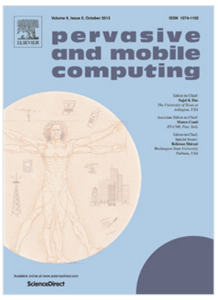
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PII: S1574-1192(26)00143-4
DOI: https://doi.org/10.1016/j.pmcj.2026.102302
Reference: PMCJ 102302

To appear in: *Pervasive and Mobile Computing*

Received date : 2 June 2026
Revised date : 15 September 2026
Accepted date : 17 October 2026

Please cite this article as: M. Javed, M.U. Ghani, H.M. Khan et al., HECATE-IDS: Explainable continual temporal graph learning for adaptive intrusion detection in IoT networks, *Pervasive and Mobile Computing* (2026), doi: https://doi.org/10.1016/j.pmcj.2026.102302.

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HECATE-IDS: Explainable Continual Temporal Graph Learning for Adaptive Intrusion Detection in IoT Networks

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***Abstract*—The rapid expansion of Internet of Things (IoT) networks has increased exposure to complex, distributed, and continuously evolving cyber threats. Although deep learning-based intrusion detection systems (IDSs) achieve high detection performance, many existing approaches rely on static traffic representations, centralized training, limited interpretability, and fixed attack distributions, reducing their effectiveness against unseen attacks, concept drift, and dynamic IoT communication behavior. This study proposes HECATE-IDS, a Hierarchical Explainable Continual Adaptive Temporal Graph Evolution framework that models IoT traffic as time-indexed cyber-behavior graphs and integrates hierarchical sparse graph-transformer learning, continual neural memory, meta-transfer drift adaptation, intrinsic graph-based threat-dependency explainability, and federated edge consensus learning. Dynamic representation captures evolving communication links, behavioral features, interaction strengths, anomaly relevance, and temporal persistence across successive observation intervals. The Explainable Causal Decision Engine provides model-intrinsic explanation evidence from sparse-attention relevance, anomaly-propagation strength, and temporally persistent graph relations without external post-hoc explainers. The framework separates local Detector Units from a central Correlation Unit, enabling local sensing, graph analysis, adaptation, explanation, and training while raw traffic remains local. HECATE-IDS is evaluated on ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023. Across five independent runs, it achieves $99.73 \pm 0.04\%$ accuracy, $99.63 \pm 0.05\%$ precision, $99.62 \pm 0.05\%$ recall, and $99.62 \pm 0.04\%$ F1-score. Under the zero-day protocol, it achieves 99.02% accuracy, 98.88% drift-adaptation performance, and a 1.34% forgetting rate. Ablation confirms complementary module contributions. Federated evaluation uses eight non-IID clients over 30 rounds, with a 4.8 MB update size and 13.7 ms average inference latency per sample. These results demonstrate statistically stable, adaptive, interpretable, and distributed intrusion detection.**

***Index Terms*—IoT intrusion detection, temporal graph learning, graph transformer, continual learning, concept drift, zero-day attack detection, explainable artificial intelligence, federated learning.**

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I. Introduction

The Internet of Things (IoT) has become a fundamental component of modern digital ecosystems, enabling intelligent services in smart homes, healthcare, industrial automation, transportation systems, smart cities, and critical infrastructures [1], [2]. The rapid growth of heterogeneous and resource-constrained IoT devices, however, has significantly expanded the cyber-attack surface. IoT networks are increasingly exposed to distributed denial-of-service (DDoS), botnet, ransomware, brute-force, spoofing, reconnaissance, injection, and advanced persistent threats [3]–[5]. Traditional security mechanisms, such as firewalls, rule-based monitoring, and signature-based intrusion detection systems (IDSs), are often inadequate for such environments because they depend on predefined attack patterns, provide limited adaptability, and struggle to detect unknown or evolving threats [6]–[8]. Machine learning and deep learning techniques have been widely explored to improve IoT intrusion detection. Classical learning models and deep neural architectures, including convolutional neural networks (CNNs), recurrent neural networks (RNNs), and long short-term memory (LSTM) networks learn discriminative traffic patterns from large-scale network data and have achieved promising detection performance [9], [10]. However, many existing IDS models still process traffic as independent feature vectors or sequential flows. Such representations are limited because IoT attacks often emerge through coordinated device interactions, abnormal communication dependencies, and evolving attack propagation paths. Moreover, conventional deep learning-based IDSs are frequently affected by high computational cost, weak interpretability, catastrophic forgetting, limited robustness under concept drift, and poor generalization to zero-day attacks [11], [12].

In this study, "dynamic traffic representation" denotes a representation that is explicitly updated across successive

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observation intervals. Rather than assuming a single fixed communication graph or independently classifying isolated records, HECATE-IDS constructs time-indexed cyber-behavior graphs whose nodes, communication links, behavioral attributes, interaction strengths, and anomaly relevance can vary as the observed traffic changes. This definition concerns the temporal representation and learning mechanism of the proposed IDS; it does not assume that every benchmark dataset is itself a continuously operating live network. The dataset-specific realization of the observation intervals and the traffic characteristics used to form evolving graphs are described in the experimental analysis.

Graph neural networks (GNNs) and graph-based learning methods have recently attracted increasing attention in cybersecurity because they represent network entities, communication links, and relational dependencies using graph structures [13]–[15]. This capability is particularly important for IoT intrusion detection, where malicious behavior may not be fully observable from isolated traffic records but can be identified through abnormal topological and temporal interactions among devices. GraphSAGE, graph attention networks, and related graph-based IDS models have shown improved performance by jointly exploiting node-level features and communication topology [16]–[20]. However, many graph-based IDS frameworks remain dependent on static graph construction and centralized model training. As a result, they cannot fully capture dynamic IoT communication behavior, preserve historical intrusion knowledge, adapt to evolving traffic distributions, or provide transparent reasoning for security analysts. A related limitation is the frequent use of a fixed attack distribution, where the attack classes, their relative occurrence, and their statistical characteristics remain approximately unchanged during training and evaluation. Such an assumption is restrictive for operational IoT networks because newly emerging attack classes, changing attack proportions, device-behavior changes, and altered communication patterns can shift the underlying data distribution. In HECATE-IDS, evolving attack distributions are addressed through continual memory and drift-aware adaptation, while zero-day conditions are represented by withholding selected attack categories from model learning and introducing them during evaluation. The exact data-partitioning and drift-evaluation procedures are specified in the experimental section to ensure reproducibility.

These limitations motivate the development of an IDS framework that jointly models temporal graph evolution, adaptive learning, explainability, and privacy-preserving distributed deployment. To address this need, this paper proposes HECATE-IDS, a Hierarchical Explainable Continual Adaptive Temporal Graph Evolution framework for IoT intrusion detection. The proposed framework transforms IoT traffic into dynamic temporal cyber-behavior graphs and learns temporal-topological intrusion representations through hierarchical sparse graph-transformer learning. It further integrates continual neural memory and meta-transfer drift adaptation to preserve previously learned threat knowledge and adapt to emerging attack patterns. Because HECATE-IDS combines several learning and deployment mechanisms, the contribution of each module must be demonstrated independently rather than inferred from the performance of the complete framework. Accordingly, the experimental design includes controlled ablation variants in which the Hierarchical Sparse Graph Transformer (HSGT), Continual Neural Memory Unit (CNMU), Meta-Transfer Drift Adapter (MTDA), Explainable Causal Decision Engine (ECDE), and Federated Edge Consensus Learning (FECL) are individually removed while the remaining training conditions are kept unchanged. The evaluation also considers computational cost relative to competing models and repeated independent runs summarized by mean performance and standard deviation so that small numerical improvements can be interpreted in terms of consistency rather than a single favorable run.

The federated IDS organization follows a local–central computational structure. At each client or edge gateway, a Detector Unit conceptually comprises a Sensor Unit and an Analyzer Unit. The Sensor Unit acquires and preprocesses local network traffic, while the Analyzer Unit performs graph construction, intrusion analysis, inference, continual adaptation, and local training using the current global model. At the central server, a Correlation Unit aggregates the model updates received from participating clients and redistributes the resulting global model for the next federated round. Thus, raw traffic remains local to the Detector Unit, whereas only model information is exchanged with the Correlation Unit. The federated evaluation therefore reports not only predictive performance but also the number of communication rounds, local-training configuration, model-update volume, aggregation latency, and communication-round cost. These measurements are used to characterize the overhead introduced by FECL. At the same time, claims regarding edge deployment are stated conservatively: the present study evaluates computational and communication feasibility on the reported experimental platform, but it does not claim validation on a dedicated low-memory edge device or physical IoT gateway. Real-device edge benchmarking is treated as a separate deployment-validation step. In addition, an explainable causal decision engine is incorporated to support interpretable intrusion reasoning and threat lineage tracing, while federated edge consensus learning enables collaborative model optimization across distributed IoT environments without sharing raw traffic data.

The proposed Explainable Causal Decision Engine (ECDE) does not employ SHAP or LIME as an external post-hoc explanation layer. Instead, it provides model-intrinsic graph-based interpretability by combining sparse graph-attention relevance, anomaly saliency, temporal dependency information, suspicious-node contribution, and estimated causal threat influence. Consequently, an intrusion decision can be related to the graph entities and communication paths that contributed most strongly to the prediction, supporting node-, link-, and attack-lineage-level interpretation.

The main contributions of this paper are summarized as follows:

- We propose HECATE-IDS, a Hierarchical Explainable Continual Adaptive Temporal Graph Evolution framework for IoT intrusion detection. Unlike conventional IDS models that rely on static traffic representations,

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HECATE-IDS models IoT traffic as time-indexed cyber-behavior graphs whose communication links, behavioral attributes, interaction strengths, and anomaly relevance evolve across successive observation intervals.

- We develop a Hierarchical Sparse Graph Transformer (HSGT) that learns multi-scale temporal-topological threat representations using anomaly-aware sparse attention. The proposed mechanism focuses computation on attack-relevant graph neighborhoods, enabling the framework to capture coordinated and multi-stage intrusion behavior while reducing the computational burden associated with unrestricted graph-transformer attention.
- We integrate a Continual Neural Memory Unit (CNMU) with a Meta-Transfer Drift Adapter (MTDA) to support continual intrusion learning, concept-drift adaptation, catastrophic-forgetting mitigation, and generalization to previously unseen malicious behavior. This mechanism preserves historical threat knowledge while adapting the learned representation to changing IoT traffic and attack distributions.
- We introduce an Explainable Causal Decision Engine (ECDE) together with Federated Edge Consensus Learning (FECL). ECDE provides intrinsic graph-based threat-dependency explanations using sparse-attention relevance, anomaly-propagation evidence, and temporal persistence, while FECL organizes distributed intrusion detection through local Detector Units and a central Correlation Unit without transferring raw client traffic. This provides a clear and reproducible local–central computational workflow for distributed IDS operation.
- We provide comprehensive experimental validation across ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023 through component-wise ablation, five independent runs with mean and standard deviation, zero-day and concept-drift evaluation, computational-cost comparison, and federated communication-overhead analysis. The evaluation further distinguishes workstation-based edge-oriented computational feasibility from unverified deployment on physical low-memory IoT gateways.

The remainder of this paper is organized as follows. Section II reviews related studies on IoT intrusion detection, graph neural networks, transformer-based IDS models, continual learning, explainable cybersecurity, and federated intrusion detection. Section III presents the proposed HECATE-IDS methodology, including dynamic behavioral graph construction, hierarchical sparse graph-transformer learning, continual neural memory, drift-aware meta-transfer adaptation, explainable causal reasoning, and federated edge consensus learning. Section IV describes the experimental setup, four benchmark IoT datasets, implementation details, evaluation metrics, and convergence analysis. Section V presents the comparative results, related-study comparison, zero-day and drift adaptation evaluation, and discussion of the obtained findings. Finally, Section VI concludes the paper and outlines future research directions.

II. Related Work

IoT intrusion detection has become an important research direction because heterogeneous and resource-constrained connected devices are increasingly exposed to DDoS attacks, botnets, spoofing, reconnaissance, injection attacks, ransomware, brute-force attempts, and other evolving cyber threats. Unlike conventional enterprise networks, IoT environments generate dynamic and heterogeneous traffic across diverse devices, protocols, and deployment conditions. As a result, traditional rule-based and signature-based IDS approaches often fail to detect unknown attacks, coordinated intrusion behavior, and rapidly changing traffic patterns. Recent studies have therefore shifted toward machine learning, deep learning, graph-based learning, transformer architectures, explainable models, and federated intrusion detection. Deep learning-based IDS models have shown strong capability in learning complex traffic patterns from large-scale IoT and wireless network data [21]. Similarly, conventional machine learning frameworks have been applied to detect IoT security attacks using feature-driven traffic analysis [22]. Optimized recurrent architectures, particularly LSTM-based IDS models, have also improved temporal traffic modeling and anomaly detection performance [23]. Although these methods provide useful detection capability, most of them process network traffic as independent feature vectors or sequential flows. This representation limits their ability to model device-to-device interactions, coordinated attack propagation, evolving communication topology, and contextual relationships among IoT entities.

Graph-based intrusion detection has emerged as a promising direction because graph neural networks can naturally represent devices, flows, communication links, and relational dependencies. Network topology-based security analysis using GNNs has demonstrated the importance of graph learning for detecting attacks that depend on structural relationships among network entities [24]. Robust GNN-based IDS studies further show that graph representations can improve detection by capturing dependencies between traffic entities and communication patterns [25]. Attribute-based graph construction has also been introduced to improve IoT intrusion detection by forming graph structures from behavioral and traffic-similarity attributes rather than relying only on fixed physical topology [26]. Other GNN-based IDS frameworks have confirmed that relational traffic modeling can enhance intrusion detection performance compared with conventional feature-based approaches [27]. In addition, recent studies on GNNs for IoT security highlight the suitability of graph learning for modeling complex IoT interactions [28]. However, many existing graph-based IDS models still rely on static graph construction and therefore cannot fully represent continuously evolving IoT communication behavior or temporal attack progression.

Attention-based graph learning has also been investigated to improve IDS performance by assigning higher importance to suspicious nodes, flows, and communication links. BS-GAT applies graph attention learning in edge-computing environments and improves intrusion detection by emphasizing attack-relevant graph relationships [29]. Graph attention-based IoT IDS models have also shown improved anomaly detection by

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learning important communication dependencies among IoT nodes [30]. Similarly, GNN-based IDS methods for industrial IoT demonstrate the usefulness of graph architectures in large-scale IIoT environments, where attacks may propagate through complex device interactions [31]. Although these studies improve topology-aware detection, they generally do not provide a unified mechanism for continual learning, drift-aware adaptation, causal explainability, and privacy-preserving distributed deployment.

Recent studies have further explored temporal attack modeling, transformer-based intrusion detection, and self-supervised graph learning. GNN-based sequential attack detection has shown that temporal traffic analysis can capture multi-stage attack behavior and evolving intrusion dependencies in IoT environments [32]. Multi-stage attack detection and prediction using GNNs also confirms the importance of modeling attack progression instead of treating each traffic instance independently [33]. Transformer-based IDS architectures, such as HiViT-IDS, improve feature representation by capturing long-range dependencies in network traffic [34]. Self-supervised GNN models have been proposed to reduce dependence on large labeled attack datasets and learn robust graph representations from unlabeled IoT traffic [35]. Nevertheless, these methods usually address temporal modeling, feature learning, or label efficiency separately, while dynamic graph evolution, catastrophic forgetting, zero-day adaptation, and explainable decision-making remain insufficiently integrated.

Explainability and robustness have also become important requirements for practical IDS deployment. Explainable CNN-based IoT IDS models have been proposed to improve analyst trust by providing interpretable evidence behind intrusion decisions [36]. Robustness-oriented studies show that GNN-based IDS models can still be vulnerable to adversarial perturbations in realistic IoT settings, emphasizing the need for more resilient graph-based detection mechanisms [37]. Graph representation learning has also been investigated to improve network intrusion detection by learning more expressive structural traffic representations [38]. In addition, heterogeneous graph neural networks combined with large language models have been explored for dual-modality intrusion detection and contextual cyberattack understanding [39]. These studies indicate a clear movement toward interpretable, robust, and context-aware IDS models. However, most explainability-oriented approaches remain post-hoc or model-specific and do not jointly address causal reasoning, continual adaptation, and distributed IoT deployment.

Interpretability mechanisms in IDSs can broadly be separated into post-hoc approaches, which explain a completed prediction, and model-intrinsic approaches, which expose decision-relevant information directly from the learned representation. HECATE-IDS follows the latter strategy. ECDE therefore differs from SHAP- or LIME-based post-hoc analysis by deriving its evidence from graph-attention relevance, anomaly saliency, temporal dependencies, and estimated causal influence over communicating entities. This design aligns the explanation process with the same temporal graph representation used for intrusion detection.

Federated and distributed intrusion detection has gained increasing attention because centralized IDS training requires collecting raw traffic data from distributed IoT devices, which may introduce privacy risks, communication overhead, and scalability limitations. Graph-based federated learning has shown that collaborative intrusion detection can be performed across distributed environments without directly sharing raw data [40]. This direction is especially relevant for IoT and edge-computing environments, where traffic is naturally distributed across devices, gateways, and local networks. However, existing federated IDS approaches mainly focus on privacy-preserving optimization and often lack temporal graph evolution, continual memory, causal explanation, and concept-drift adaptation.

For multi-component IDS architectures, high aggregate accuracy alone is insufficient to establish that every integrated module is necessary. Component-wise ablation is therefore important for separating the contribution of representation learning, memory retention, drift adaptation, explanation, and federated optimization. Likewise, increasingly complex graph-transformer and federated models should be evaluated not only in terms of classification performance but also in terms of computational and communication cost. When performance margins between competing IDS methods are small, repeated independent runs and dispersion statistics are additionally required to distinguish consistent improvement from run-to-run variation. These considerations motivate the ablation, efficiency, communication-overhead, and repeated-run analyses included in the experimental evaluation of HECATE-IDS.

For federated intrusion detection, architectural clarity also requires separating the local detection functions from the central aggregation function. In the proposed organization, the local Detector Unit performs sensing, preprocessing, analysis, prediction, and local training, while the central Correlation Unit performs federated aggregation and global-model redistribution. This separation provides a direct mapping between conventional IDS functions and the federated learning workflow.

To make the distinction between HECATE-IDS and representative prior studies explicit, Table I summarizes the major capabilities emphasized in the reviewed literature.

The reviewed literature shows that IoT intrusion detection has progressed from conventional feature-based learning toward graph-aware, attention-driven, temporal, explainable, and federated models. However, several important gaps remain unresolved. First, many deep learning-based IDS models achieve high detection accuracy but do not explicitly capture the relational structure of IoT communication. Second, most graph-based IDS methods depend on static graph construction and therefore cannot fully model evolving attack behavior. Third, temporal and transformer-based IDS models improve representation learning but usually lack continual memory and privacy-preserving distributed deployment. Fourth, explainable IDS methods improve interpretability but often do not address zero-day adaptation, concept drift, or catastrophic forgetting. Finally, federated IDS approaches support privacy-preserving learning but generally lack causal reasoning and dynamic temporal graph intelligence.

The taxonomy shows that the novelty of HECATE-IDS

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TABLE I
COMPARATIVE TAXONOMY OF REPRESENTATIVE IOT INTRUSION DETECTION STUDIES

Study	Primary Approach	Traffic / Graph Representation	Temporal Learning	Continual / Drift Adaptation	Explainability	Federated Learning	Main Gap Relative to HECATE-IDS
Ngo et al. [26]	Attribute-based GNN IDS	Attribute-derived graph	Limited	No	No	No	Improves graph construction but does not jointly address temporal evolution, continual adaptation, explainability, and federated learning.
Wang et al. [29]	Graph Attention Network IDS	Graph-based	Limited	No	No	No	Emphasizes attack-relevant graph relationships but does not integrate drift-aware continual learning or federated collaboration.
Altaf et al. [32]	Sequential attack detection using GNNs	Graph-based sequential traffic	Yes	No	No	No	Models sequential attack behavior but provides limited support for continual adaptation, explainability, and distributed federated training.
Zhou et al. [34]	Hierarchical Vision Transformer IDS	Feature/sequence representation	Partial	No	No	No	Captures long-range feature dependencies but does not explicitly construct evolving cyber-behavior graphs.
Ebrahimi et al. [36]	Explainable CNN-based IoT IDS	Feature-based	No	No	Yes	No	Improves interpretability but does not integrate temporal graph evolution, continual learning, or federated deployment.
Al Tfaily [40]	Graph-based federated IDS	Graph-based	Partial	No	Limited	Yes	Supports privacy-preserving collaborative learning but provides limited integration of temporal evolution, drift adaptation, and causal explanation.
HECATE-IDS	**Hierarchical sparse temporal graph transformer with continual, explainable, federated learning**	**Dynamic time-indexed cyber-behavior graphs**	**Yes**	**Yes**	**Yes**	**Yes**	**Jointly integrates dynamic graph modeling, temporal learning, continual memory, drift adaptation, intrinsic causal explanation, and federated edge consensus.**

does not arise from the isolated use of a graph neural network, transformer, continual learner, explanation module, or federated optimizer. Instead, its principal contribution is the integration of these capabilities into one time-evolving IDS workflow: dynamic behavioral graph construction represents changing communication relationships; HSGT learns temporal-topological threat representations; CNMU preserves previously acquired threat knowledge; MTDA supports adaptation under distributional change; ECDE provides intrinsic graph-based explanations; and FECL enables distributed optimization without transferring raw client traffic. This integrated design positions HECATE-IDS against prior studies that address these requirements individually or in smaller combinations. Quantitative comparison with baseline and state-of-the-art methods is provided in the results section.

These limitations motivate the proposed HECATE-IDS framework, which integrates dynamic behavioral graph construction, hierarchical sparse graph-transformer learning, continual neural memory, meta-transfer drift adaptation, explainable causal decision-making, and federated edge consensus learning into a unified framework for adaptive and interpretable IoT intrusion detection. Accordingly, the evaluation of HECATE-IDS is structured to answer two complementary questions: whether each integrated component contributes measurable value, and whether the resulting accuracy gains remain justified when statistical variability and execution cost are considered. The present study therefore separates algorithmic effectiveness from deployment claims; measured resource and communication results are used to discuss feasibility, while validation on dedicated low-memory edge hardware is not assumed without direct experimentation.

III. PROPOSED METHODOLOGY

This section presents the proposed HECATE-IDS framework, a Hierarchical Explainable Continual Adaptive Temporal Graph Evolution model for IoT intrusion detection. The framework is designed to address the main limitations of existing IDS approaches, including static traffic representation, weak zero-day attack adaptation, limited interpretability, catastrophic forgetting, concept drift, and scalability constraints in distributed IoT environments. Unlike conventional IDS models that process network traffic as independent feature vectors or sequential flows, HECATE-IDS represents IoT communication as dynamic temporal cyber-behavior graphs and learns evolving intrusion patterns through sparse graph-transformer learning, continual memory, drift-aware adaptation, explainable causal reasoning, and federated edge consensus optimization. The proposed architecture consists of six major components: Dynamic Behavioral Graph Constructor (DBGC), Hierarchical Sparse Graph Transformer (HSGT), Continual Neural Memory Unit (CNMU), Meta-Transfer Drift Adapter (MTDA), Explainable Causal Decision Engine (ECDE), and Federated Edge Consensus Learning (FECL). These components operate jointly to capture temporal-topological intrusion dependencies, preserve historical cyber-threat knowledge, adapt to evolving traffic distributions, provide interpretable threat-dependency reasoning, and support privacy-preserving distributed deployment.

To quantify the contribution of each major module rather than attributing the final performance only to their joint integration, the experimental analysis includes a controlled component-wise ablation protocol. Five reduced variants are evaluated under the same data partitions, optimization settings, and evaluation metrics as the complete HECATE-IDS model: (i) HECATE-IDS without HSGT, in which hierarchical sparse

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graph-transformer processing is bypassed and the graph-ready representation is forwarded directly to the downstream adaptive modules; (ii) HECATE-IDS without CNMU, where historical threat-memory updates are disabled; (iii) HECATE-IDS without MTDA, where drift-sensitive parameter adaptation is removed; (iv) HECATE-IDS without ECDE, where the explanation-specific causal evidence and its associated regularization term are disabled while the prediction path is retained; and (v) HECATE-IDS without FECL, where the model is evaluated without federated global aggregation. This ablation design isolates the marginal contribution of representation learning, continual memory, drift adaptation, intrinsic explainability, and federated optimization. The quantitative results of these variants are reported in the Results and Discussion section.

In the proposed framework, *dynamic traffic representation* refers to the construction of a sequence of time-indexed cyber-behavior graphs rather than the use of a single fixed graph or independent static traffic records. For each observation interval τ, the active entities, communication links, behavioral features, interaction strengths, and anomaly relevance can change according to the traffic observed during that interval. Thus, temporal evolution is represented explicitly through the sequence $\{G_1, G_2, \ldots, G_T\}$ and is further exploited by continual memory and drift-aware adaptation across successive intervals. This definition separates the notion of dynamic representation from the assumption that a benchmark dataset must itself be a continuously operating live network stream; recorded benchmark traffic can be organized into ordered observation intervals and represented as evolving graphs during model processing.

The federated IDS architecture is organized into two computational levels. At the local/client level, each edge client operates a *Detector Unit*, which contains a *Sensor Unit* and an *Analyzer Unit*. The Sensor Unit acquires local traffic and performs preprocessing and temporal-window formation. The Analyzer Unit executes DBGC, HSGT, CNMU, MTDA, and ECDE for local graph construction, intrusion analysis, prediction, adaptation, explanation, and local model optimization. At the central level, a *Correlation Unit* performs FECL-based aggregation of the client model updates and redistributes the updated global model. Raw traffic remains at the local Detector Units; only model information is communicated to the Correlation Unit.

Algorithm 1 summarizes the complete learning process of HECATE-IDS. At each time interval, IoT traffic is transformed into a dynamic cyber-behavior graph. Each edge client extracts temporal-topological representations, updates continual memory, estimates drift intensity, generates intrinsic graph-based explanations, and optimizes the local objective. The local model updates are then aggregated through federated edge consensus learning to update the global IDS model without sharing raw traffic data.

To make the federated IDS computation explicit, Algorithm 2 separately summarizes the interaction between the local Detector Units and the central Correlation Unit. The two algorithms are complementary: Algorithm 1 describes the learning operations of HECATE-IDS, whereas Algorithm 2 clarifies where these operations are executed in the distributed IDS architecture.

Algorithm 1: HECATE-IDS Learning Algorithm

Input: IoT traffic stream X^τ, edge clients $\mathcal{E}$, global model Θ_g, memory bank $\mathcal{M}$
Output: Predicted class $\hat{y}^\tau$, probability P^τ, explanation score $\mathcal{X}^\tau$
Components: DBGC, HSGT, CNMU, MTDA, ECDE, FECL
Hyperparameters: α, β, η, μ_τ, ζ, λ_1, λ_2, λ_3, λ_4
Initialize Θ_g^0, $\{\Theta_e^0\}_{e=1}^{|\mathcal{E}|}$, and $\mathcal{M}^0$
for $\tau = 1$ **to** T **do**
  foreach *client* $e \in \mathcal{E}$ **do**
    Receive global model $\Theta_g^{\tau-1}$ from the central Correlation Unit
    `// Sensor Unit: acquire and preprocess local traffic`
    $X_e^\tau \leftarrow \mathrm{SensorProcess}(X^\tau, e)$
    `// Analyzer Unit: construct the local dynamic cyber-behavior graph`
    $G_{e,\tau} = (N_{e,\tau}, L_{e,\tau}, F_{e,\tau})$
    $\omega_{ij}^{e,\tau} = \alpha \Psi_{ij}^{e,\tau} + \beta \Gamma_{ij}^{e,\tau}$
    Extract local graph embedding Z_e^τ
    Compute sparse graph attention
    $\Lambda_{ij}^\tau = \mathrm{softmax}_{j \in \mathcal{N}(i)} \left(\frac{Q_i^\tau (K_j^\tau)^T \Omega_{ij}^\tau}{\sqrt{d}} \right)$
    Learn temporal-topological representation
    $H_e^\tau = \mathrm{HSGT}(Z_e^\tau, \Lambda^\tau)$
    Update continual memory
    $\mathcal{M}^\tau = \eta \mathcal{M}^{\tau-1} + (1-\eta)\Xi^\tau$
    Estimate concept drift
    $\Delta_\tau = \left\| \Phi_\tau^{src} - \Phi_\tau^{tgt} \right\|_2$
    Update drift-adaptive parameters
    $\Theta_e^\tau = \Theta_e^{\tau-1} - \mu_\tau \nabla \mathcal{J}_\tau$
    `// Compute ECDE threat-dependency explanation evidence`
    $\chi_{ij}^\tau = \frac{1}{h} \sum_{s=\max(1,\tau-h+1)}^{\tau} \mathbb{I}[(i,j) \in L_s]$
    $\widetilde{\Gamma}_{ij}^\tau = \mathrm{Normalize}\left(\Gamma_{ij}^\tau\right)$
    $\Pi_{ij}^\tau = \widetilde{\Gamma}_{ij}^\tau \chi_{ij}^\tau$
    $R_{ij}^\tau = \Lambda_{ij}^\tau \Pi_{ij}^\tau$
    $\mathcal{X}_\tau = 100 \frac{\sum_{(i,j) \in L_\tau} R_{ij}^\tau}{\sum_{(i,j) \in L_\tau} \Lambda_{ij}^\tau + \epsilon}$
    Generate prediction probability
    $P^\tau = \mathrm{softmax}(W_c H_e^\tau + b_c)$
    $\hat{y}^\tau = \arg\max(P^\tau)$
    Compute hybrid loss
    $\mathcal{L}_e^\tau = \lambda_1 \mathcal{L}_{cls} + \lambda_2 \mathcal{L}_{drift} + \lambda_3 \mathcal{L}_{mem} + \lambda_4 \mathcal{L}_{causal}$
    Update local model
    $\Theta_e^\tau \leftarrow \Theta_e^\tau - \zeta \nabla_{\Theta_e} \mathcal{L}_e^\tau$
    Send the local model update to the central Correlation Unit
  end
  `// Correlation Unit: aggregate local model updates`
  $\Theta_g^\tau = \frac{1}{|\mathcal{E}|} \sum_{e=1}^{|\mathcal{E}|} \Theta_e^\tau$
  Broadcast Θ_g^τ to all edge clients
end
return $\hat{y}^\tau$, P^τ, $\mathcal{X}^\tau$

Fig. 1 presents the architecture of the proposed HECATE-IDS framework. The framework first performs traffic processing, where raw IoT traffic is converted into feature-level representations, temporal windows, and graph-ready data. The processed traffic is then passed to the Dynamic Behavioral Graph Constructor (DBGC), which models IoT communication behavior using nodes, links, temporal weights, and anomaly-related connections. The resulting cyber-behavior graph is processed by the Hierarchical Sparse Graph Transformer (HSGT), which learns discriminative threat represen-

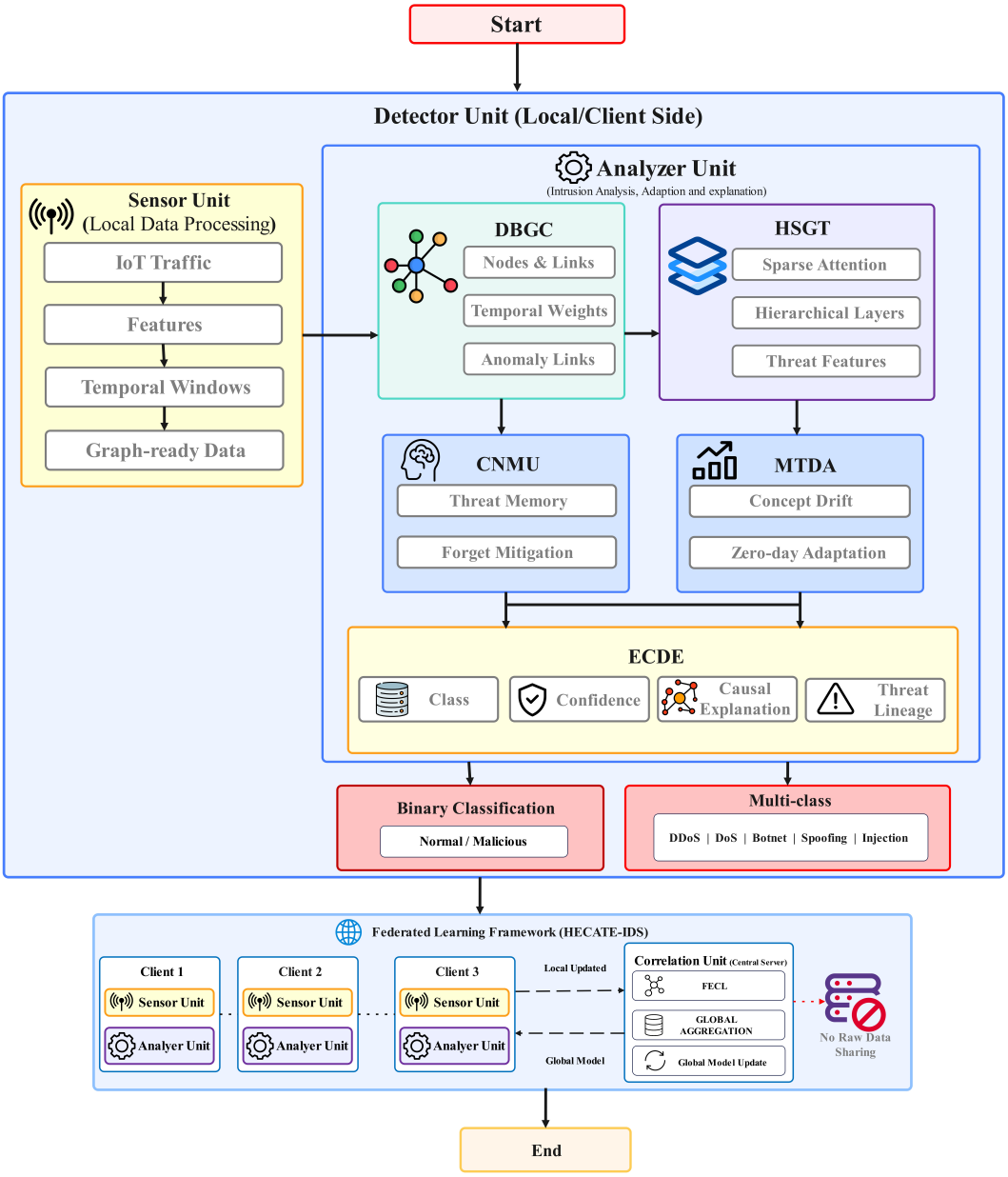


Fig. 1. Architecture of the proposed HECATE-IDS framework.

tations through sparse attention, hierarchical graph layers, and temporal-topological feature extraction.

The learned representations are refined through two adaptive learning modules: the Continual Neural Memory Unit (CNMU) and the Meta-Transfer Drift Adapter (MTDA). CNMU preserves historical threat knowledge and reduces catastrophic forgetting, while MTDA supports concept-drift handling and zero-day attack adaptation. The refined representations are then forwarded to the Explainable Causal Decision Engine (ECDE), which generates intrusion decisions together with confidence estimation, causal explanation, and threat lineage information. The final detection layer supports both binary classification and multi-class attack identification. In parallel, Federated Edge Consensus Learning (FECL) enables distributed model optimization through local updates and global aggregation without sharing raw traffic data, thereby supporting scalable and privacy-preserving deployment in heterogeneous IoT environments.

From an IDS-network perspective, the computational path in Fig. 1 can be interpreted as follows. First, each edge client or gateway hosts a Detector Unit. Its Sensor Unit receives local traffic, performs preprocessing, and organizes traffic into ordered observation intervals. Second, its Analyzer Unit executes DBGC, HSGT, CNMU, MTDA, and ECDE to construct the current graph, learn threat representations, maintain historical knowledge, detect drift, classify traffic, and

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Algorithm 2: Federated Detector–Correlation Workflow

Input: Local client traffic $\{X_e\}_{e=1}^{|\mathcal{E}|}$, initial global model Θ_g^0
Output: Updated global model and local intrusion decisions
for *each federated communication round* r **do**
  Central Correlation Unit broadcasts Θ_g^{r-1} to participating clients
  foreach *client* $e \in \mathcal{E}$ **do**
    `// Local Detector Unit`
    Sensor Unit collects local traffic and forms ordered temporal windows
    Sensor Unit preprocesses the current window and generates graph-ready features
    Analyzer Unit constructs $G_{e,\tau}$ using DBGC
    Analyzer Unit extracts sparse temporal-topological features using HSGT
    Analyzer Unit updates continual knowledge using CNMU
    Analyzer Unit estimates distributional drift using MTDA
    Analyzer Unit predicts the intrusion class and produces ECDE explanation evidence
    Analyzer Unit performs local optimization using the hybrid objective
    Send only the local model update Θ_e^r to the Correlation Unit
  end
  `// Central Correlation Unit`
  Aggregate $\{\Theta_e^r\}$ using FECL global consensus averaging
  Produce updated global model Θ_g^r
end
return Θ_g^r *and local intrusion predictions*

generate explanation evidence. Third, local model optimization is performed without transmitting the raw traffic outside the client. Fourth, the central Correlation Unit receives only the model updates, performs FECL aggregation, and redistributes the global parameters for the next communication round. This local–central separation clarifies both the IDS functionality and the federated learning computation.

A. Dynamic Behavioral Graph Constructor

Conventional IDS models usually represent IoT traffic as tabular features or sequential packet flows. Although these representations are useful for basic traffic classification, they do not fully capture the relational structure of IoT networks, where attacks often emerge through coordinated communication among devices. Botnet coordination, lateral movement, stealth communication, and distributed attack propagation are inherently relational and temporal. Therefore, the proposed Dynamic Behavioral Graph Constructor transforms graph-ready IoT traffic into dynamic cyber-behavior graphs that capture both device-level interactions and evolving intrusion dependencies.

Dynamic behavior is represented by partitioning the observed traffic into ordered intervals and constructing one graph for each interval. Consequently, HECATE-IDS does not assume that network topology, communication intensity, or anomaly relevance remain constant. A communication relation can appear, disappear, strengthen, or weaken between two consecutive graphs as the traffic changes. This enables recorded benchmark traffic to be processed as an evolving sequence while preserving the temporal order used by the framework.

At each temporal interval τ, the IoT environment is represented as:

$$G_\tau = (N_\tau, L_\tau, F_\tau) \tag{1}$$

where N_τ denotes the set of IoT device nodes, L_τ represents communication links, and F_τ denotes node-edge behavioral features. Unlike static graph construction, the proposed graph is updated over time according to changes in traffic behavior, communication frequency, temporal interaction patterns, and anomaly intensity.

Accordingly, the dynamic representation can be expressed as the ordered graph sequence

$$\mathcal{G}_{1:T} = \{G_1, G_2, \ldots, G_T\}, \tag{2}$$

where a change from $G_{\tau-1}$ to G_τ reflects an observed change in the communication structure and/or behavioral attributes. This sequence provides the temporal context used by the subsequent continual-learning and drift-adaptation modules.

The interaction strength between nodes n_i and n_j is defined as:

$$\omega_{ij}^\tau = \alpha \Psi_{ij}^\tau + \beta \Gamma_{ij}^\tau \tag{3}$$

where Ψ_{ij}^τ represents normalized communication intensity, Γ_{ij}^τ denotes anomaly propagation strength, and α and β are adaptive balancing coefficients. This formulation enables HECATE-IDS to assign higher importance to suspicious communication links and capture evolving intrusion propagation patterns in large-scale IoT environments.

Therefore, "dynamic IoT communication behavior" in this work refers to changes in graph connectivity, edge interaction strength, behavioral features, and anomaly relevance across observation intervals. It should be distinguished from network-system performance quantities such as communication latency, bandwidth consumption, or energy usage, which concern deployment overhead rather than traffic-behavior representation.

B. Hierarchical Sparse Graph Transformer

After constructing the temporal cyber-behavior graph, HECATE-IDS employs a Hierarchical Sparse Graph Transformer (HSGT) to learn temporal-topological intrusion representations. Standard transformer attention requires each node to attend to all other nodes, resulting in quadratic computational complexity, which is inefficient for large IoT graphs with massive device interactions. To reduce this cost, HSGT applies anomaly-aware sparse attention, where each node attends only to selected neighbors with high attack relevance.

Let the graph embedding matrix at time interval τ be:

$$Z_\tau \in \mathbb{R}^{m \times d} \tag{4}$$

where m denotes the number of nodes and d represents the latent embedding dimension. The sparse graph attention score between node n_i and node n_j is computed as:

$$\Lambda_{ij}^\tau = \frac{\exp\left(\frac{(Q_i^\tau)(K_j^\tau)^T \Omega_{ij}^\tau}{\sqrt{\delta}}\right)}{\sum_{k \in M(i)} \exp\left(\frac{(Q_i^\tau)(K_k^\tau)^T \Omega_{ik}^\tau}{\sqrt{\delta}}\right)} \tag{5}$$

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where Q and K denote query and key projections, δ is the scaling dimension, Ω_{ij}^{τ} represents anomaly saliency, and $M(i)$ denotes the selected sparse neighborhood of node n_i. The hierarchical structure enables lower layers to capture local communication anomalies, while higher layers model broader attack propagation across IoT sub-networks. This design improves the detection of coordinated and multi-stage intrusions while reducing unnecessary attention computation.

The temporal component is obtained by applying HSGT over the ordered graph sequence and passing the learned representations to the continual-memory and drift-adaptation stages. Hence, HSGT captures the structural dependencies within each observation interval, while CNMU and MTDA preserve and adapt information across successive intervals. This division avoids interpreting a single graph snapshot as the complete temporal model.

C. Continual Neural Memory Unit

IoT attack behavior changes over time as new devices, communication patterns, vulnerabilities, and attack strategies emerge. IDS models trained on fixed traffic distributions may therefore lose previously learned intrusion knowledge when adapting to new threats, resulting in catastrophic forgetting. To address this limitation, HECATE-IDS introduces a Continual Neural Memory Unit (CNMU) that stores representative cyber-threat patterns and updates them progressively during learning.

The episodic memory bank at time interval τ is defined as:

$$\mathcal{M}_{\tau} = \{(u_1, \rho_1), (u_2, \rho_2), \ldots, (u_p, \rho_p)\} \tag{6}$$

where u_p denotes a latent attack embedding and ρ_p represents its corresponding threat significance score. The memory bank stores compact representations of previously observed intrusion behavior, anomaly signatures, temporal fingerprints, and attack propagation patterns.

The memory update operation is formulated as:

$$\mathcal{M}_{\tau} = \eta \mathcal{M}_{\tau-1} + (1-\eta)\Xi_{\tau} \tag{7}$$

where $\mathcal{M}_{\tau-1}$ denotes retained historical knowledge, Ξ_{τ} represents newly acquired intrusion knowledge at time interval τ, and η controls the stability-plasticity trade-off. A larger value of η preserves previous threat knowledge, whereas a smaller value enables faster adaptation to emerging attack patterns. This mechanism allows HECATE-IDS to retain historical intrusion knowledge while continuously adapting to zero-day and evolving cyber threats.

The role of CNMU in evolving attack distributions is therefore to maintain continuity between historical and current threat representations. When newly observed traffic differs from previous intervals, the current representation Ξ_{τ} is incorporated without completely replacing the historical state. This mechanism directly supports the study's treatment of non-stationary attack behavior and mitigates catastrophic forgetting during adaptation.

D. Meta-Transfer Drift Adapter

IoT traffic distributions are highly dynamic because of device heterogeneity, topology changes, firmware updates, user behavior variation, and evolving adversarial strategies. As a result, IDS models trained on historical traffic may experience performance degradation when deployed under new or shifted distributions. To address this issue, HECATE-IDS employs a Meta-Transfer Drift Adapter (MTDA), which estimates distributional changes and updates model representations according to the observed drift intensity.

At time interval τ, the concept drift magnitude is estimated by measuring the distance between the source-domain and target-domain latent representations:

$$\Delta_{\tau} = \left\| \Phi_{\tau}^{\text{src}} - \Phi_{\tau}^{\text{tgt}} \right\|_2 \tag{8}$$

where Φ_{τ}^{src} and Φ_{τ}^{tgt} denote the latent embeddings of the source and target traffic distributions, respectively. A higher value of Δ_{τ} indicates stronger deviation between previously learned traffic behavior and the currently observed IoT traffic distribution.

Operationally, Φ_{τ}^{src} represents the reference latent behavior learned from previously observed traffic, whereas Φ_{τ}^{tgt} represents the latent behavior of the current observation interval. The quantity Δ_{τ} therefore provides a model-space measure of distributional change. This formulation connects the notion of a fixed attack distribution to the proposed adaptive mechanism: if the traffic distribution remains stable, Δ_{τ} remains small; when attack composition or behavioral characteristics change, the latent distance increases and triggers stronger adaptation through the drift-sensitive update coefficient.

The drift-aware adaptation process updates the model parameters as follows:

$$\Theta_{\tau}^{*} = \Theta_{\tau-1} - \mu_{\tau} \nabla J_{\tau} \tag{9}$$

where Θ_{τ}^{*} represents the updated drift-adaptive model parameters, $\Theta_{\tau-1}$ denotes the previous model state, μ_{τ} is a drift-sensitive adaptation coefficient, and ∇J_{τ} is the gradient of the drift-adaptation objective. This update enables HECATE-IDS to respond to evolving attack distributions while maintaining stable intrusion detection performance under concept drift.

E. Explainable Causal Decision Engine

High detection accuracy alone is insufficient for practical IDS deployment because security analysts require interpretable evidence behind model predictions. Many deep learning and graph-based IDS models operate as black-box classifiers, making it difficult to identify why a device, flow, or communication path is classified as malicious. To improve transparency, HECATE-IDS incorporates an Explainable Causal Decision Engine (ECDE), which combines sparse graph-attention relevance, anomaly-propagation strength, and temporal graph persistence to generate intrinsic graph-based explanation evidence.

ECDE does not employ SHAP, LIME, or another external post-hoc explanation method. Instead, it reuses quantities already generated by the HECATE-IDS learning pipeline.

Specifically, Λ_{ij}^{τ} represents the sparse-attention relevance of the communication relation between nodes i and j, Γ_{ij}^{τ} represents the anomaly-propagation strength produced during dynamic graph construction, and the ordered graph sequence provides information about whether a suspicious relation persists across recent observation intervals. Consequently, the explanation can be traced directly to communication relations that are simultaneously important to the detector, anomalous in the current graph, and temporally persistent.

To operationalize the previously described threat-influence quantity, the temporal persistence of relation (i, j) at interval τ is defined as

$$\chi_{ij}^{\tau} = \frac{1}{h} \sum_{s=\max(1,\tau-h+1)}^{\tau} \mathbb{I}\left[(i,j) \in L_s\right], \tag{10}$$

where h denotes the number of recent observation intervals considered, $\mathbb{I}[\cdot]$ is the indicator function, and $\chi_{ij}^{\tau} \in [0, 1]$ measures how consistently the relation appears across the recent dynamic graph sequence.

The anomaly-propagation quantity Γ_{ij}^{τ} is min–max normalized within the current graph as

$$\widetilde{\Gamma}_{ij}^{\tau} = \frac{\Gamma_{ij}^{\tau} - \Gamma_{\min}^{\tau}}{\Gamma_{\max}^{\tau} - \Gamma_{\min}^{\tau} + \epsilon}, \tag{11}$$

where $\epsilon > 0$ is a small numerical-stability constant. The estimated threat-influence score is then computed as

$$\Pi_{ij}^{\tau} = \widetilde{\Gamma}_{ij}^{\tau} \chi_{ij}^{\tau}, \tag{12}$$

so that a relation receives high threat influence only when it exhibits strong anomaly propagation and remains relevant across recent temporal graph states.

The contribution of relation (i, j) to the explanation is obtained by combining the learned sparse-attention relevance with the estimated threat influence:

$$R_{ij}^{\tau} = \Lambda_{ij}^{\tau} \Pi_{ij}^{\tau}. \tag{13}$$

The graph-level threat-dependency score is therefore

$$\Upsilon_{\tau} = \sum_{(i,j)\in L_{\tau}} R_{ij}^{\tau} = \sum_{(i,j)\in L_{\tau}} \Lambda_{ij}^{\tau} \Pi_{ij}^{\tau}. \tag{14}$$

Relations with larger R_{ij}^{τ} values are ranked as more influential evidence for the current intrusion decision, enabling the explanation to identify suspicious nodes, links, and temporally persistent communication paths.

To obtain a bounded and reproducible explanation-quality measure, the Explainability Confidence Score (ECS) is defined as the attention-weighted average threat influence:

$$\mathcal{X}_{\tau} = 100 \frac{\sum_{(i,j)\in L_{\tau}} \Lambda_{ij}^{\tau} \Pi_{ij}^{\tau}}{\sum_{(i,j)\in L_{\tau}} \Lambda_{ij}^{\tau} + \epsilon}. \tag{15}$$

Because $\Pi_{ij}^{\tau} \in [0, 1]$ and the sparse-attention weights are non-negative, $\mathcal{X}_{\tau}$ is bounded to $[0, 100]$ and is reported directly as a percentage. The dataset-level ECS used in the experimental analysis is obtained by averaging $\mathcal{X}_{\tau}$ over the evaluated prediction intervals. A higher ECS therefore indicates stronger agreement between learned attention relevance, anomaly-propagation evidence, and temporal persistence.

The ECDE explanation procedure can be summarized in four stages: (i) identify attack-relevant graph relations using Λ_{ij}^{τ}; (ii) quantify temporally persistent anomaly influence using Π_{ij}^{τ}; (iii) rank relation-level explanation contributions R_{ij}^{τ}; and (iv) associate the highest-ranked relations with the predicted intrusion class and its temporal threat lineage. The term "causal" in the component name denotes model-internal threat-dependency reasoning over temporally connected cyber interactions. It is not intended to claim formal intervention-based causal identification in the statistical or structural-causal-model sense. This distinction preserves the intended threat-lineage interpretation while avoiding an unsupported causal-inference claim.

Because ECS is generated from ECDE-specific quantities, it is defined only when the ECDE explanation pathway is present. Consequently, the ECDE ablation does not produce an ECS value. Any cross-model ECS comparison must therefore use an equivalent explanation pathway; otherwise, ECS should be reported as not applicable for models that do not contain ECDE.

F. Federated Edge Consensus Learning

Centralized IDS training requires collecting raw traffic from distributed IoT devices and edge gateways, which may introduce privacy risks, communication overhead, and scalability bottlenecks. To mitigate these issues, HECATE-IDS employs Federated Edge Consensus Learning (FECL), where each edge client trains a local IDS model using private traffic data and shares only model updates for global aggregation. This enables collaborative intrusion learning without exposing sensitive network traffic.

Within the IDS architecture, the local FECL participant corresponds to the Detector Unit. The Sensor Unit never sends raw traffic to the server; instead, the Analyzer Unit trains the local HECATE-IDS model on the client-side traffic partition. The central FECL server corresponds to the Correlation Unit, whose role is model correlation and consensus rather than direct packet inspection. This mapping makes the federated learning roles consistent with conventional IDS terminology.

Each edge client updates its local model parameters as follows:

$$\Theta_e^{\tau} = \Theta_e^{\tau-1} - \zeta \nabla L_e^{\tau} \tag{16}$$

where Θ_e^{τ} denotes the local model parameters of edge client e at time interval τ, ζ is the local learning rate, and L_e^{τ} represents the local training loss.

The global consensus model is obtained through federated aggregation:

$$\Theta_{\text{global}}^{\tau} = \frac{1}{|E|} \sum_{e=1}^{|E|} \Theta_e^{\tau} \tag{17}$$

where $|E|$ denotes the number of participating edge clients. The aggregated global model is redistributed to local clients for the next communication round. This strategy supports

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privacy-preserving and edge-oriented distributed IDS operation while reducing dependence on centralized traffic collection. The term "edge-oriented" is used deliberately: the present study evaluates federated execution, communication overhead, latency, and resource consumption on the reported experimental platform, but it does not claim validation on a physical low-memory IoT gateway or embedded edge board. Consequently, real-device deployment on platforms such as resource-constrained gateways remains an important direction for future hardware-level evaluation.

The present methodological formulation focuses on the learning and privacy-preserving computation flow of FECL. Dynamic IoT communication behavior is captured through the changing graph structure and traffic behavior processed by each Detector Unit; it should not be interpreted as a claim that communication latency, bandwidth overhead, energy consumption, or hardware utilization are modeled by Eqs. (16)–(17). Those deployment-level quantities require a separate systems evaluation and are therefore treated independently from the intrusion-detection representation and learning mechanism.

G. Computational Complexity and Communication-Cost Formulation

HECATE-IDS integrates multiple learning and adaptation modules, computational efficiency is analyzed explicitly rather than inferred only from classification accuracy. Let m denote the number of graph nodes in an observation interval, $|L_\tau|$ the number of retained communication links, d the latent embedding dimension, p the number of stored memory prototypes, K the number of participating federated clients, and P the number of trainable model parameters. A conventional dense graph-transformer attention layer requires pairwise node interactions with a dominant complexity of approximately $\mathcal{O}(m^2 d)$. In contrast, HSGT restricts attention to the selected sparse neighborhood and therefore reduces the dominant attention computation to approximately $\mathcal{O}(|L_\tau| d)$ for each sparse-attention layer, excluding implementation-dependent projection and feed-forward costs.

The additional adaptive modules introduce lower-order operations relative to dense all-pairs attention. CNMU updates and retrieves compact latent threat representations with a cost that scales with the maintained memory size, approximately $\mathcal{O}(pd)$ for a direct comparison against p stored representations. MTDA operates on source and target latent representations and introduces drift-estimation and gradient-update costs that scale primarily with the latent and trainable parameter dimensions. ECDE reuses sparse-attention relevance, anomaly-propagation strength, and temporal graph-presence information already available from the preceding stages; its threat-influence computation and relation-ranking operations scale with the retained graph links and the temporal persistence horizon rather than with unrestricted all-pairs node interactions. These theoretical expressions are complemented by the empirical runtime, memory, latency, and model-size comparisons reported in the experimental section.

For federated execution, communication cost is also analyzed explicitly. If S_{upd} denotes the serialized size of one client model update and S_{global} denotes the size of the redistributed global model, the aggregate bidirectional communication volume for one round with K participating clients can be expressed as

$$C_{\mathrm{round}} = K\left(S_{\mathrm{upd}} + S_{\mathrm{global}}\right), \tag{18}$$

while the total communication volume over R global rounds is

$$C_{\mathrm{total}} = R\, C_{\mathrm{round}}. \tag{19}$$

This formulation separates privacy preservation from communication efficiency: FECL avoids transferring raw traffic, but model exchange still incurs measurable communication overhead. Accordingly, the experimental analysis reports the number of communication rounds, model-update size, client upload/download volume, aggregation latency, and end-to-end round latency.

H. Global Optimization Objective

The proposed HECATE-IDS framework is optimized using a unified objective function that jointly considers intrusion classification, drift adaptation, continual memory preservation, and ECDE explanation consistency:

$$L_{\text{total}} = \lambda_1 L_{\text{cls}} + \lambda_2 L_{\text{drift}} + \lambda_3 L_{\text{memory}} + \lambda_4 L_{\text{causal}} \tag{20}$$

where L_{cls} is the intrusion classification loss, L_{drift} guides adaptation to changing traffic distributions, L_{memory} preserves historical intrusion knowledge, and L_{causal} denotes the ECDE explanation-consistency regularizer associated with stable threat-dependency evidence. The coefficients λ_1, λ_2, λ_3, and λ_4 control the contribution of each objective during training.

The four terms have complementary roles under evolving traffic conditions: L_{cls} maintains discriminative intrusion detection, L_{drift} penalizes inadequate adaptation to distributional change, L_{memory} discourages loss of previously learned threat knowledge, and L_{causal} encourages stable ECDE explanation evidence for the graph relations supporting the prediction rather than imposing a formal causal-identification objective.

The final intrusion decision is obtained from the predicted probability distribution:

$$\hat{y}_\tau = \arg\max\left(P_\tau\right) \tag{21}$$

where P_τ denotes the output probability distribution at time interval τ. The prediction layer supports both binary and multi-class intrusion detection. In the binary setting, the model distinguishes benign and malicious traffic, whereas in the multi-class setting it identifies fine-grained attack categories such as DDoS, DoS, botnet, spoofing, injection, reconnaissance, ransomware, and brute-force attacks.

The integration of dynamic graph construction, sparse graph-transformer learning, continual memory, drift-aware adaptation, intrinsic threat-dependency explainability, and federated edge learning enables HECATE-IDS to provide an adaptive, interpretable, and scalable IDS framework for next-generation IoT cybersecurity environments.

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The proposed methodology therefore addresses the architectural and interpretability questions at the model-design level: evolving communication behavior is represented through time-indexed graphs, non-stationary attack behavior is handled through CNMU and MTDA, explainability is provided intrinsically by ECDE through attention-weighted temporal threat-dependency evidence, and federated IDS computation is explicitly divided between local Detector Units and a central Correlation Unit. The experimental section separately specifies how zero-day exposure, client-wise attack distributions, concept drift, and communication-round evaluation are instantiated in the reported experiments.

IV. Experimental Analysis

This section evaluates the effectiveness of the proposed HECATE-IDS framework under diverse IoT intrusion detection scenarios. The experiments are designed to assess classification performance, convergence behavior, robustness, scalability, explainability, continual adaptation, concept-drift handling, and zero-day detection capability. The proposed framework is evaluated under both binary and multi-class classification settings. In the binary setting, traffic samples are classified as benign or malicious, whereas in the multi-class setting, the model identifies fine-grained attack categories such as DDoS, DoS, reconnaissance, spoofing, botnet, injection, ransomware, brute-force, and other IoT-specific attacks. The ordinary binary and multi-class evaluations use attack categories available during supervised learning. The zero-day experiment is evaluated separately: one complete attack category per dataset is excluded from all training clients and is introduced only after federated training as previously unseen malicious traffic. This separation prevents the zero-day result from being conflated with supervised multi-class recognition.

To strengthen the empirical validation of the integrated architecture, the revised evaluation additionally includes three complementary analyses. First, a component-wise ablation study removes HSGT, CNMU, MTDA, ECDE, and FECL individually to quantify their respective contributions. Second, all comparative models are evaluated over five independent runs with different random seeds, and the principal performance metrics are reported as mean $\pm$ standard deviation. Third, computational and federated communication costs are measured to assess the additional overhead introduced by combining multiple learning, adaptation, explanation, and aggregation components. To clarify the meaning of dynamic evaluation in this study, the benchmark traffic is processed as ordered observation intervals from which time-indexed cyber-behavior graphs are constructed. Dynamic behavior is represented through changes in graph connectivity, communication intensity, behavioral features, anomaly relevance, and attack composition across successive intervals. The datasets are therefore used as temporally ordered traffic sources from which evolving graph states are generated, rather than being treated as independent static records.

The experiments were conducted on four benchmark IoT intrusion detection datasets: ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023. These datasets provide heterogeneous traffic distributions, diverse attack types, protocol-specific IoT traffic, botnet-oriented attack behavior, and large-scale IoT intrusion scenarios, making them suitable for evaluating adaptive IoT intrusion detection. Due to the large scale of ToN-IoT, BoTIoT, and CICIoT2023, stratified mini-batch training and class-preserving sampling were used where necessary to ensure computational feasibility while maintaining representative class distributions. All experiments were executed on an HP ZBook 15 workstation equipped with an Intel Core i7 processor, 32 GB DDR4 RAM, an NVIDIA RTX A1000 GPU with 6 GB dedicated memory, and 512 GB NVMe SSD storage. The implementation was developed using Python 3.10, TensorFlow, PyTorch Geometric, CUDA 12.0, and Scikit-learn.

The federated clients in this study are simulated logical IoT edge clients on the reported workstation rather than independent physical low-memory gateways. Therefore, the present experiments evaluate edge-oriented federated execution through client-wise training, communication overhead, latency, memory consumption, and model-update exchange, but they do not constitute direct validation on embedded hardware such as Raspberry Pi- or Jetson-class gateways. Accordingly, claims regarding edge suitability are restricted to the measured computational profile, while real-device validation is retained as future work.

A. Federated Experimental Configuration

The federated evaluation was performed using eight simulated IoT edge clients/gateways. Federated training was conducted for 30 global communication rounds. During each round, every participating client performed five local epochs using a mini-batch size of 32 before transmitting its model update to the central Correlation Unit. All eight clients participated in every communication round, corresponding to a client participation ratio of 100%. The global model was initialized once and redistributed to the participating clients after each aggregation round. For communication-cost analysis, the serialized size of each local model update, the aggregate client upload per round, the global-model download size, server-side aggregation latency, and end-to-end communication-round latency were recorded throughout the federated execution. These measurements are reported separately from classification performance so that the privacy benefit of keeping raw traffic local is evaluated together with the communication burden introduced by model exchange.

To represent heterogeneous client traffic conditions, local data partitions were generated using a non-IID Dirichlet distribution with concentration parameter $\alpha = 0.5$. This configuration creates heterogeneous attack proportions among clients, such that individual Detector Units observe different combinations and frequencies of benign and malicious traffic while contributing to a common federated model. The resulting setup more closely reflects distributed IoT environments in which traffic characteristics and attack exposure differ across gateways.

Table II summarizes the federated learning setup used in HECATE-IDS. The evaluation uses eight non-IID clients, 30 communication rounds, five local epochs per round, and 100%

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client participation, while raw traffic remains local and only model updates are shared with the central Correlation Unit.

TABLE II
FEDERATED LEARNING CONFIGURATION USED IN HECATE-IDS

Parameter	Configuration
Number of Edge Clients	8
Global Communication Rounds	30
Local Epochs per Round	5
Client Participation Ratio	100%
Batch Size	32
Client Distribution	Non-IID Dirichlet, $\alpha = 0.5$
Aggregation	Global consensus averaging
Raw-data Sharing	Not allowed
Repeated Runs	5 independent executions
Communication Metrics	Update size, upload/download volume, aggregation latency, communication-exchange latency

B. Component-Wise Ablation Protocol

A controlled ablation study was conducted to determine whether each major module contributes measurably to the complete framework. The full HECATE-IDS configuration was used as the reference, and one component was removed at a time while preserving the same data partitions, optimization conditions, client configuration, and evaluation metrics wherever applicable. The five ablation variants were defined as follows: HECATE-IDS without HSGT, without CNMU, without MTDA, without ECDE, and without FECL.

Table III summarizes the component-wise ablation configurations used to evaluate the contribution of HSGT, CNMU, MTDA, ECDE, and FECL. Each component is removed individually while the remaining experimental settings are kept unchanged.

The ablation variants were evaluated using classification accuracy, F1-score, zero-day accuracy, drift-adaptation performance, forgetting rate, and Explainability Confidence Score (ECS), where applicable. ECS is generated from ECDE-specific quantities and is therefore not reported for the variant without ECDE because the intrinsic explanation pathway is absent. The corresponding quantitative ablation results are presented in the Results and Discussion section.

C. Repeated-Run Statistical Evaluation

To assess the stability of the reported improvements, each implemented model was executed five times using different random seeds while maintaining the same preprocessing, train–test partitions, and principal optimization settings. For a metric x, the mean over the five independent runs is computed as

$$\bar{x} = \frac{1}{5}\sum_{r=1}^{5} x_r, \tag{22}$$

and the sample standard deviation is computed as

$$s_x = \sqrt{\frac{1}{4}\sum_{r=1}^{5}(x_r - \bar{x})^2}, \tag{23}$$

where x_r denotes the metric obtained in run r. Final comparative results are therefore reported in the form $\bar{x} \pm s_x$. This repeated-run protocol is particularly important because the absolute performance differences between high-performing IDS models are relatively small; the standard deviation provides direct evidence regarding whether the observed ranking remains stable across independent executions.

D. Computational-Cost Evaluation Protocol

HECATE-IDS integrates graph representation learning, continual memory, drift adaptation, explainability, and federated optimization, its computational cost was evaluated explicitly. The analysis records model-update size, average local inference latency, throughput, average local-training time per communication round, peak system-memory utilization, peak GPU-memory utilization, average CPU/GPU utilization, and communication latency. For the implemented competing models, computational measurements are collected under the same workstation and software environment wherever the model architecture permits a direct comparison.

Federated communication overhead is evaluated separately from local computation. With K participating clients, serialized client-update size S_{upd}, and redistributed global-model size S_{global}, the bidirectional communication volume per round is

$$C_{\mathrm{round}} = K\left(S_{\mathrm{upd}} + S_{\mathrm{global}}\right), \tag{24}$$

and over R global communication rounds,

$$C_{\mathrm{total}} = R\, C_{\mathrm{round}}. \tag{25}$$

The empirical communication measurements reported in the Results and Discussion section complement this formulation and quantify the practical cost of FECL model exchange.

E. Dataset Description

Table IV summarizes the benchmark datasets used in this study. The selected datasets cover heterogeneous IoT telemetry, botnet traffic, MQTT protocol-based attacks, and large-scale IoT threat scenarios. This diversity enables a comprehensive evaluation of HECATE-IDS across modern IoT cybersecurity environments.

The suitability of these datasets for dynamic traffic modeling arises from their behavioral diversity. ToN-IoT provides heterogeneous IoT telemetry and multiple attack behaviors; BoTIoT contains botnet-oriented traffic with multiple malicious activities; MQTTIoT provides protocol-specific communication among interconnected sensors; and CICIoT2023 contains traffic from 105 IoT devices under diverse attack scenarios. These characteristics enable HECATE-IDS to form ordered temporal windows and capture variations in communication and attack behavior across graph states.

1) ToN-IoT Dataset: The ToN-IoT dataset represents large-scale IoT telemetry and network traffic collected from heterogeneous IoT environments. It includes normal traffic and multiple attack categories, such as Scanning, XSS, DDoS, Password, DoS, Injection, Backdoor, MITM, and Ransomware attacks. As summarized in Table V, the dataset contains 16,977,496 samples with 43 features. Its scale and attack diversity make it suitable for evaluating temporal graph learning,

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TABLE III
COMPONENT-WISE ABLATION CONFIGURATIONS USED IN THE EVALUATION

Configuration	Disabled Component	Evaluation Objective
Full HECATE-IDS	None	Reference configuration containing HSGT, CNMU, MTDA, ECDE, and FECL.
HECATE-IDS w/o HSGT	Hierarchical Sparse Graph Transformer	Quantifies the contribution of sparse temporal-topological graph representation learning.
HECATE-IDS w/o CNMU	Continual Neural Memory Unit	Quantifies the effect of historical threat-memory preservation and forgetting mitigation.
HECATE-IDS w/o MTDA	Meta-Transfer Drift Adapter	Quantifies the contribution of drift-sensitive adaptation under changing traffic distributions.
HECATE-IDS w/o ECDE	Explainable Causal Decision Engine	Quantifies the effect of ECDE-specific threat-dependency explanation and explanation-consistency regularization while retaining the prediction path.
HECATE-IDS w/o FECL	Federated Edge Consensus Learning	Quantifies the effect of collaborative global aggregation across heterogeneous clients.

TABLE IV
SUMMARY OF BENCHMARK DATASETS USED IN THE PROPOSED STUDY

Dataset	Features	Samples	Attack Categories	Dynamic/Behavioral Relevance
ToN-IoT	43	16,977,496	Scanning, XSS, DDoS, Password, DoS, Injection, Backdoor, MITM, Ransomware	Heterogeneous IoT telemetry and diverse attack behaviors support time-indexed behavioral graph construction.
BoTIoT	46	4,292,000	DDoS, DoS, Reconnaissance, Keylogging, OS Scan, Service Scan, Data Exfiltration	Botnet-oriented traffic and multiple attack activities support changes in attack behavior and propagation patterns.
MQTTIoT	23	495,134	Flood, SlowITe, Malformed Data, DoS, BruteForce	Protocol-specific MQTT traffic from interconnected sensors supports temporal analysis of communication anomalies.
CICIoT2023	46	46,686,579	DDoS, DoS, Mirai, Spoofing, Reconnaissance, Web-based, BruteForce	Large-scale traffic from 105 IoT devices and diverse attacks supports heterogeneous and evolving graph states.

adaptive intrusion detection, and robustness under heterogeneous IoT traffic distributions.

For dynamic graph construction, the heterogeneous traffic records are organized into ordered observation intervals. Variations in communicating entities, traffic intensity, feature behavior, and attack labels across these intervals provide the information used by DBGC to update successive cyber-behavior graphs.

TABLE V
DESCRIPTION OF THE TON-IOT DATASET

Property	Value
Number of Features	43
Number of Samples	16,977,496
Number of Classes	Normal + 9 attack classes
Normal Samples	6,099,469
Scanning Samples	3,781,419
XSS Samples	2,455,020
DDoS Samples	2,026,234
Password Samples	1,153,323
DoS Samples	712,609
Injection Samples	684,465
Backdoor Samples	16,809
MITM Samples	7,723
Ransomware Samples	3,425

2) BoTIoT Dataset: The BoTIoT dataset is designed for evaluating botnet-oriented IoT intrusion detection. It contains benign and malicious traffic generated in a simulated IoT botnet environment. The attack categories include DDoS, DoS, Reconnaissance, Keylogging, OS Scan, Service Scan, and Data Exfiltration. Since the original dataset is extremely large, a representative class-preserving subset was used for efficient experimentation and graph construction. Table VI reports the dataset configuration used in this study.

The coexistence of benign traffic, high-volume denial-of-service behavior, reconnaissance activity, scanning, and theft-related attacks provides distinct behavioral regimes from which the framework can construct changing graph states and evaluate adaptation to different attack patterns.

TABLE VI
DESCRIPTION OF THE BOTIOT DATASET

Property	Value
Number of Features	46
Number of Samples	4,292,000
Number of Classes	Normal + 7 attack classes
Normal Samples	477,000
DDoS Samples	1,300,000
DoS Samples	1,100,000
Reconnaissance Samples	1,000,000
Theft Samples	50,000
Keylogging Samples	15,000
OS Scan Samples	200,000
Service Scan Samples	150,000

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3) MQTTIoT Dataset: The MQTTIoT dataset evaluates intrusion detection under MQTT protocol-based IoT communication. It contains traffic generated from a realistic MQTT-enabled environment with multiple interconnected sensors. The dataset includes Flood, SlowITe, Malformed Data, DoS, and BruteForce attacks. As shown in Table VII, it contains 495,134 samples with 23 features, making it useful for evaluating protocol-aware temporal intrusion detection and MQTT-specific anomaly behavior.

Because the dataset is centered on MQTT communication among interconnected sensors, changes in protocol-level traffic behavior across ordered intervals provide a focused setting for evaluating whether HECATE-IDS can capture temporal communication anomalies rather than relying only on static feature values.

TABLE VII
DESCRIPTION OF THE MQTTIOT DATASET

Property	Value
Number of Features	23
Number of Samples	495,134
Number of Classes	Normal + 5 attack classes
Normal Samples	330,204
Flood Samples	58,356
SlowITe Samples	44,864
Malformed Data Samples	35,295
DoS Samples	16,455
BruteForce Samples	9,960

4) CICIoT2023 Dataset: The CICIoT2023 dataset is a large-scale benchmark for modern IoT intrusion detection. It contains network traffic generated from 105 IoT devices under diverse attack scenarios, including DDoS, DoS, Mirai, Spoofing, Reconnaissance, Web-based attacks, and BruteForce attacks. As shown in Table VIII, the dataset includes 46,686,579 records with 46 features. Its scale and attack diversity provide a strong testbed for evaluating the robustness and scalability of HECATE-IDS under large-scale IoT attack conditions.

The presence of traffic from 105 IoT devices and several attack families provides the strongest multi-device setting among the four datasets for evaluating changes in communication relations, attack composition, and anomaly intensity across time-indexed graph states.

TABLE VIII
DESCRIPTION OF THE CICIOT2023 DATASET

Property	Value
Number of Features	46
Number of Samples	46,686,579
Number of Classes	Normal + 7 attack classes
Benign Samples	1,098,195
DDoS Samples	33,984,560
DoS Samples	8,090,738
Mirai Samples	2,634,124
Spoofing Samples	486,504
Recon Samples	354,565
Web Samples	24,829
BruteForce Samples	13,064

F. Zero-Day and Concept-Drift Evaluation Protocol

The zero-day and drift experiments separate ordinary supervised recognition from adaptation to previously unseen or shifted attack behavior. Let $\mathcal{C}$ denote the complete set of attack categories and $\mathcal{C}_u \subset \mathcal{C}$ the attack category selected as unseen. Samples belonging to $\mathcal{C}_u$ are excluded from all federated client training partitions and introduced only during testing. Specifically, Ransomware is held out for ToN-IoT, Theft for BoTIoT, BruteForce for MQTTIoT, and Web-based attacks for CICIoT2023. This design guarantees that none of the eight participating clients receives training samples from the corresponding zero-day class before evaluation. Exactly one attack category is withheld for each dataset. The held-out category is removed before the remaining training samples are partitioned among the eight clients using the non-IID Dirichlet distribution. Consequently, the unseen class is absent from every local client dataset throughout all 30 communication rounds and cannot enter the global model indirectly through another client. Only after federated training is completed are samples from the corresponding held-out class introduced for zero-day testing. In this zero-day experiment, the held-out samples are evaluated as previously unseen malicious behavior rather than as a supervised multi-class label learned during training. Therefore, zero-day accuracy measures whether the trained detector can distinguish unseen malicious traffic from benign traffic; it does not imply that the model learned the semantic label of an attack category that was completely absent from training.

Let TP_u denote held-out attack samples correctly detected as malicious, FN_u denote held-out attack samples incorrectly classified as benign, TN_b denote benign samples correctly classified as benign, and FP_b denote benign samples incorrectly classified as malicious. The zero-day accuracy is computed as

$$\mathrm{Acc}_{ZD} = \frac{TP_u + TN_b}{TP_u + FN_u + TN_b + FP_b}, \tag{26}$$

while the corresponding zero-day detection rate is

$$\mathrm{ZDR} = \frac{TP_u}{TP_u + FN_u}. \tag{27}$$

Accordingly, the reported zero-day result quantifies detection of attack behavior that was completely unavailable during model training. Multi-class attack identification reported elsewhere in the manuscript is evaluated separately using attack categories available during supervised learning.

Concept drift is simulated over three ordered traffic phases. Phase 1 follows the original training distribution. In Phase 2, the relative proportion of selected attack classes is increased by 20%, while Phase 3 introduces a stronger distribution shift of 35%. At each interval τ, the drift magnitude is computed as $\Delta_\tau = \|\Phi_\tau^{src} - \Phi_\tau^{tgt}\|_2$, where Φ_τ^{src} is the reference latent representation and Φ_τ^{tgt} is the representation of the current traffic interval. The average latent drift magnitude decreases from 0.47 before adaptation to 0.18 after MTDA adaptation. Correspondingly, classification accuracy recovers from 91.6% immediately after the induced drift to 98.9% after adaptation.

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TABLE IX
ZERO-DAY CLASS CONFIGURATION USED IN THE FEDERATED EVALUATION

Dataset	Held-Out Attack	Training Exposure	Testing Condition
ToN-IoT	Ransomware	Excluded from all clients	Introduced only during zero-day malicious-vs-benign testing
BoTIoT	Theft	Excluded from all clients	Introduced only during zero-day malicious-vs-benign testing
MQTTIoT	BruteForce	Excluded from all clients	Introduced only during zero-day malicious-vs-benign testing
CICIoT2023	Web-based attack	Excluded from all clients	Introduced only during zero-day malicious-vs-benign testing

Algorithm 3: Zero-Day and Concept-Drift Evaluation Protocol

Input: Ordered traffic intervals, client set $\mathcal{E}$, class set $\mathcal{C}$, held-out class set $\mathcal{C}_u$, global model Θ_g
Output: Binary zero-day predictions, drift scores, adapted global model
Remove all samples in $\mathcal{C}_u$ from federated training data
Create non-IID client partitions using Dirichlet $\alpha = 0.5$
for $r = 1$ **to** *30* **do**
  Broadcast current global model Θ_g^{r-1} to all 8 clients
  foreach *client* $e \in \mathcal{E}$ **do**
    Process the current ordered traffic interval
    Construct the local cyber-behavior graph $G_{e,\tau}$
    Extract the current latent representation Φ_τ^{tgt}
    Obtain the historical/reference representation Φ_τ^{src}
    Compute $\Delta_\tau = \|\Phi_\tau^{src} - \Phi_\tau^{tgt}\|_2$
    Update CNMU memory and adapt the model through MTDA
    Perform 5 local epochs using batch size 32
    Send only the local model update to the Correlation Unit
  end
  Aggregate local updates using FECL to obtain Θ_g^r
end
Introduce the held-out class set $\mathcal{C}_u$ only during zero-day testing and treat its samples as malicious
Evaluate Acc_{ZD}, ZDR, post-adaptation accuracy, and forgetting rate
return *zero-day predictions,* Δ_τ *values, adapted global model*

Algorithm 3 summarizes the zero-day and concept-drift evaluation procedure. The held-out attack class is excluded from all federated training clients and introduced only during testing, while drift adaptation is performed through CNMU and MTDA during the 30 communication rounds.

G. Evaluation Metrics

The performance of HECATE-IDS is evaluated using Accuracy, Precision, Recall, F1-score, Detection Rate (DR), False Positive Rate (FPR), and Explainability Confidence Score (ECS). These metrics provide a comprehensive evaluation of classification correctness, attack detection capability, false-alarm behavior, and explanation quality.

Accuracy measures the proportion of correctly classified samples:

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN} \tag{28}$$

where TP, TN, FP, and FN denote true positives, true negatives, false positives, and false negatives, respectively.

Precision measures the proportion of predicted attack samples that are truly malicious:

$$Precision = \frac{TP}{TP + FP} \tag{29}$$

Recall measures the proportion of actual attack samples correctly detected by the model:

$$Recall = \frac{TP}{TP + FN} \tag{30}$$

The F1-score provides a balanced measure between Precision and Recall:

$$F1 = \frac{2 \times Precision \times Recall}{Precision + Recall} \tag{31}$$

The Detection Rate measures the ability of the model to correctly identify malicious traffic:

$$DR = \frac{TP}{TP + FN} \tag{32}$$

The False Positive Rate measures the proportion of benign samples incorrectly classified as attacks:

$$FPR = \frac{FP}{FP + TN} \tag{33}$$

The Explainability Confidence Score (ECS) evaluates the internal consistency of the explanation generated by ECDE.

In accordance with the ECDE formulation in Section III, ECS is computed from the attention-weighted threat influence associated with the retained graph relations. For each interval τ, the threat-influence quantity Π_{ij}^τ combines normalized anomaly-propagation strength and temporal persistence, while Λ_{ij}^τ represents sparse graph-attention relevance. The interval-level ECS is therefore

$$\mathcal{X}_\tau = 100 \frac{\sum_{(i,j)\in L_\tau} \Lambda_{ij}^\tau \Pi_{ij}^\tau}{\sum_{(i,j)\in L_\tau} \Lambda_{ij}^\tau + \epsilon}, \tag{34}$$

and the dataset-level ECS is obtained by averaging $\mathcal{X}_\tau$ across the evaluated prediction intervals. Because the score is generated from ECDE-specific quantities, ECS is reported only when the ECDE explanation pathway is present. It is therefore not applicable to the HECATE-IDS variant without ECDE, and it should not be interpreted as a generic explanation metric for models that do not implement an equivalent ECDE pathway. A higher ECS indicates stronger agreement among learned attention relevance, anomaly-propagation evidence, and temporal persistence. For the comparative and ablation experiments,

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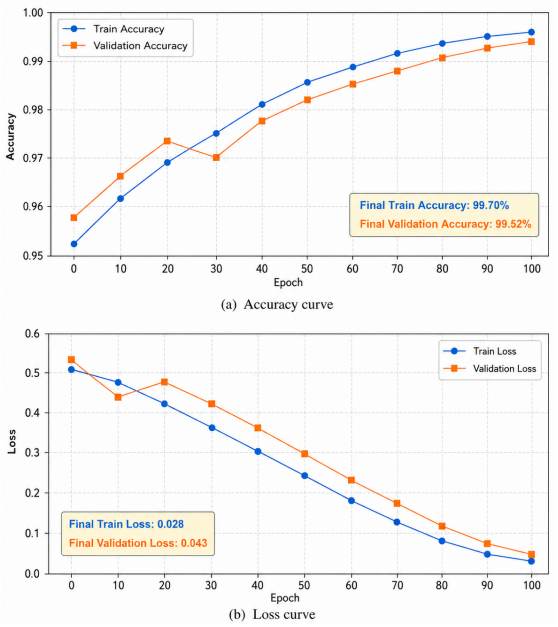


Fig. 2. Training and validation accuracy/loss curves for the ToN-IoT dataset.

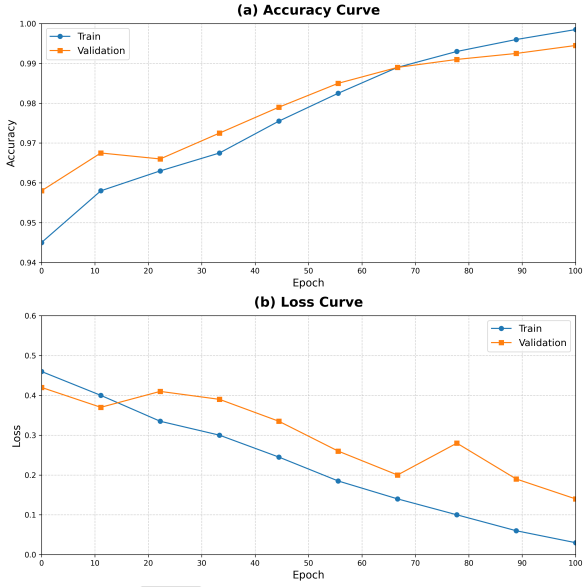


Fig. 3. Training and validation accuracy/loss curves for the BoTIoT dataset.

the principal classification metrics are reported as mean ± standard deviation over five independent runs according to Eqs. (22) and (23). This statistical reporting allows small absolute differences between high-performing models to be interpreted together with their run-to-run variability.

H. Training Convergence Analysis

Training convergence was analyzed using accuracy and loss curves across the benchmark IoT datasets. The proposed HECATE-IDS framework was trained for 100 epochs, and convergence behavior was monitored using both training and validation performance. This analysis is important because high detection accuracy must be supported by stable optimization behavior, decreasing validation loss, and a small train-validation gap. The 100-epoch curves are model-level convergence diagnostics and are distinct from the federated communication schedule. The federated experiments use 30 global communication rounds with five local epochs per client in each round. Thus, epoch-wise optimizer convergence and communication-round convergence are reported as separate aspects of the experimental analysis.

Fig. 2 shows the training and validation accuracy/loss curves for the ToN-IoT dataset. The accuracy curves increase steadily and converge toward stable performance, while the loss curves decrease consistently across epochs. HECATE-IDS achieves a final training accuracy of 99.70% and validation accuracy of 99.52%, with final training and validation losses of 0.028 and 0.043, respectively. The small gap between training and validation performance indicates stable generalization and limited overfitting on large-scale heterogeneous IoT traffic.

Fig. 3 presents the convergence behavior on the BoTIoT dataset. The accuracy curves show gradual and stable improvement, while the loss curves consistently decrease during training. The close alignment between training and validation curves suggests that the model learns generalized botnet behavior patterns without severe overfitting. Minor validation-loss fluctuations are expected because of heterogeneous and imbalanced botnet traffic, but the downward trend confirms stable optimization.

V. Results and Discussion

This section presents the comparative evaluation and discussion of the proposed HECATE-IDS framework. The evaluation was conducted using four benchmark IoT intrusion detection datasets, namely ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023, under binary and multi-class classification settings. These datasets were selected because they represent large-scale heterogeneous IoT traffic, botnet-oriented attacks, MQTT protocol-specific threats, and modern large-scale IoT intrusion scenarios. The proposed framework was trained for 100 epochs, selected empirically based on validation convergence and drift-adaptation behavior. Unlike conventional sequential models, HECATE-IDS uses dynamic temporal cyber-behavior graphs and anomaly-aware sparse graph attention to capture evolving communication patterns and attack dependencies. The 100-epoch setting reported for model-level convergence is distinct from the federated training schedule, which consists of 30 global communication rounds with five local epochs per participating client in each round. In addition to the single-configuration comparisons, the revised analysis reports five-run mean ± standard deviation, component-wise ablation, zero-day and drift behavior, and federated computational/communication overhead to quantify both performance consistency and the contribution of the integrated modules. The zero-day results reported in this section refer to unseen malicious-versus-benign detection using attack categories completely excluded from training, whereas ordinary multi-class results refer to attack categories available during supervised learning.

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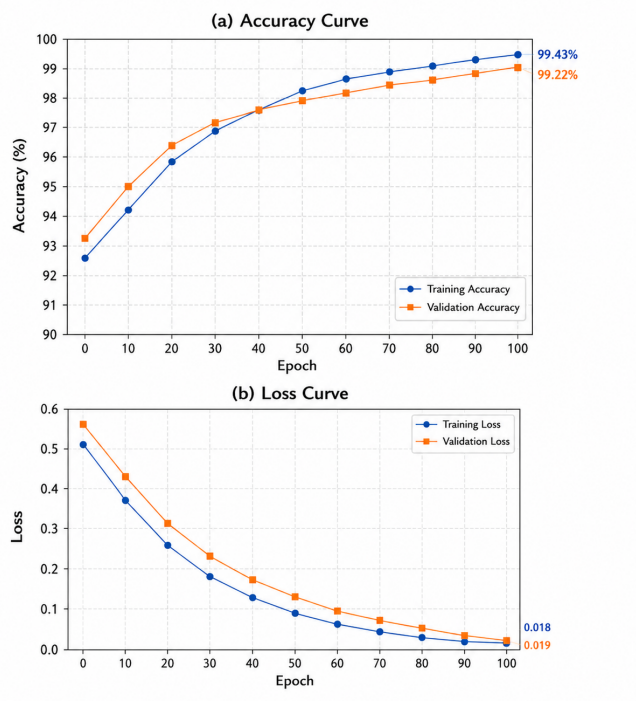


Fig. 4. Accuracy and loss curves of the proposed HECATE-IDS framework for the MQTTIoT dataset.

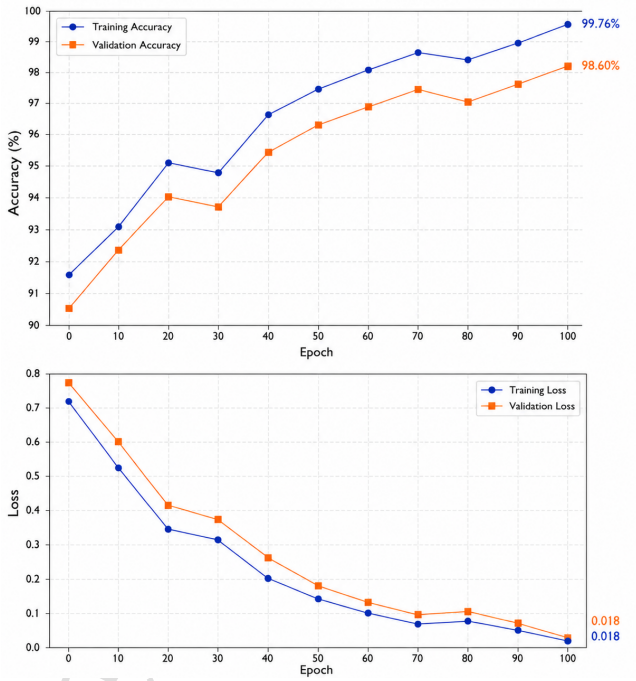


Fig. 5. Accuracy and loss curves of the proposed HECATE-IDS framework for the CICIoT2023 dataset.

Fig. 4 illustrates the learning behavior of HECATE-IDS on the MQTTIoT dataset. The accuracy curves converge steadily, while the loss curves decrease consistently during training.

Fig. 5 shows stable convergence on CICIoT2023 despite its scale and heterogeneous attack distribution.

A. Comparison with Existing Models

The proposed HECATE-IDS framework was compared with Autoencoders, CNN, RNN, LSTM, GraphSAGE, GAT, and Swin-LSTM using consistent benchmark datasets and the reported evaluation metrics.

B. Comparison with Published State-of-the-Art Studies

In addition to the controlled comparison with the implemented baseline models, the performance of HECATE-IDS is further contextualized against representative state-of-the-art intrusion detection studies reported in the recent literature. This comparison is included to clarify the effectiveness and novelty of the proposed framework beyond the architectural taxonomy presented in the Related Work section. Because the compared studies employ different dataset variants, preprocessing procedures, train–test partitions, class definitions, and evaluation protocols, the reported values are used as literature-level reference points rather than as a strictly controlled head-to-head ranking.

Table XI shows that recent specialized IDS models can achieve very high classification performance, particularly under binary detection settings. Therefore, the contribution of HECATE-IDS should not be interpreted solely in terms of obtaining the highest individual accuracy value. Its primary distinction is the unified treatment of dynamic temporal graph representation, continual threat-memory preservation, concept-drift adaptation, intrinsic graph-based explainability, zero-day evaluation, and federated model optimization. For example, HiViT-IDS reports 99.70% accuracy on ToN-IoT [34], whereas HECATE-IDS achieves 99.81% on the same benchmark under the experimental configuration used in this study. Similarly, Altaf et al. [32] report 99.97% binary accuracy on BoT-IoT, confirming that near-saturated binary performance is achievable by specialized temporal graph models. Consequently, the practical significance of HECATE-IDS lies not only in its consistently high classification results, but also in its ability to combine adaptation, knowledge retention, interpretability, zero-day robustness, and distributed learning within a single framework. Because the published studies differ in preprocessing, data partitions, dataset variants, and evaluation metrics, the numerical differences in Table XI are interpreted as contextual evidence rather than as controlled pairwise improvements.

C. Dataset-Wise Performance Analysis

Tables XII, XIII, XIV, and XV present the performance comparison of HECATE-IDS with the baseline models on ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023, respectively. HECATE-IDS achieves the highest overall performance across all four benchmark datasets.

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TABLE X
PARAMETER SETTINGS OF THE COMPARED MODELS

Parameter	Autoencoders	CNN	RNN	LSTM	GraphSAGE	GAT	Swin-LSTM	HECATE-IDS
Number of Layers	3	5	3	4	4	6	8	8 hierarchical graph layers
Hidden Units	64	128	100	128	128	256	3×96	12 sparse graph heads
Activation Function	ReLU	ReLU	tanh	tanh	ReLU	GELU	GELU	GELU
Optimizer	Adam	Adam	Adam	Adam	AdamW	AdamW	AdamW	AdamW
Learning Rate	0.001	0.001	0.001	0.0005	0.0005	0.0003	0.0003	0.0001
Batch Size	32	32	32	32	16	16	16	32
Loss Function	CE	CE	CE	CE	CE	CE	CE	Hybrid adaptive loss
Epochs	50	50	50	75	75	100	100	100

TABLE XI
PERFORMANCE COMPARISON WITH REPRESENTATIVE PUBLISHED STATE-OF-THE-ART IOT INTRUSION DETECTION STUDIES

Study	Method	Dataset(s)	Reported Performance	Comparison Context
Wang et al. [29]	BS-GAT	NF-BoT-IoT-v2 and NF-ToN-IoT-v2	Binary precision, recall, F1-score, and accuracy exceed 99%; multi-class accuracy exceeds 93%	Demonstrates strong graph-attention-based intrusion detection, but does not jointly incorporate continual memory, explicit concept-drift adaptation, intrinsic graph-based threat-dependency explanation, and federated learning.
Altaf et al. [32]	Sequential GGCN	BoT-IoT and Mirai	99.97% binary accuracy on BoT-IoT; 98.86% multi-class accuracy on BoT-IoT; 76.70% multi-class accuracy on Mirai	Demonstrates effective sequential graph learning and temporal botnet detection, but does not integrate the full combination of continual adaptation, federated learning, and intrinsic graph-based explainability considered in HECATE-IDS.
Zhou et al. [34]	HiViT-IDS	ToN-IoT and Edge-IIoTset	99.70% accuracy on ToN-IoT and 100% accuracy on Edge-IIoTset	Provides a directly relevant transformer-based benchmark. HECATE-IDS achieves 99.81% accuracy on ToN-IoT while additionally incorporating dynamic graph representation, continual memory, drift adaptation, intrinsic explanation, and federated learning.
Al Tfaily et al. [40]	FedGATSage	NF-ToN-IoT and CIC-ToN-IoT	78.58% balanced accuracy on NF-ToN-IoT and 80.24% balanced accuracy on CIC-ToN-IoT	Provides a graph-based federated IDS reference. Because balanced accuracy and different dataset variants are used, the reported values are included for contextual comparison rather than direct numerical ranking.
HECATE-IDS	**Hierarchical sparse temporal graph transformer with continual, explainable, and federated learning**	**ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023**	**99.73% average accuracy, 99.62% average F1-score, and 99.02% zero-day accuracy**	**Jointly evaluates dynamic temporal graph representation, continual memory, concept-drift adaptation, intrinsic graph-based explainability, zero-day robustness, and non-IID federated learning across four IoT datasets.**

TABLE XII
PERFORMANCE COMPARISON ON THE TON-IOT DATASET

Model	Accuracy	Precision	Recall	F1-score
Autoencoders	95.12%	94.84%	94.63%	94.73%
CNN	96.84%	96.21%	96.03%	96.12%
RNN	97.52%	97.10%	96.88%	96.99%
LSTM	98.44%	98.12%	98.03%	98.07%
GraphSAGE	98.87%	98.55%	98.43%	98.49%
GAT	99.04%	98.91%	98.82%	98.86%
Swin-LSTM	99.27%	99.12%	99.03%	99.07%
HECATE-IDS	**99.81%**	**99.74%**	**99.88%**	**99.81%**

TABLE XIII
PERFORMANCE COMPARISON ON THE BOTIOT DATASET

Model	Accuracy	Precision	Recall	F1-score
Autoencoders	94.91%	94.37%	94.21%	94.29%
CNN	96.55%	96.11%	95.84%	95.97%
RNN	97.42%	97.01%	96.88%	96.94%
LSTM	98.27%	98.01%	97.93%	97.97%
GraphSAGE	98.91%	98.63%	98.55%	98.59%
GAT	99.11%	98.94%	98.83%	98.88%
Swin-LSTM	99.39%	99.17%	99.04%	99.10%
HECATE-IDS	**99.92%**	**99.83%**	**99.79%**	**99.81%**

TABLE XIV
PERFORMANCE COMPARISON ON THE MQTTIOT DATASET

Model	Accuracy	Precision	Recall	F1-score
Autoencoders	93.44%	92.88%	92.55%	92.71%
CNN	95.76%	95.24%	94.91%	95.07%
RNN	96.88%	96.41%	96.03%	96.22%
LSTM	97.94%	97.62%	97.33%	97.47%
GraphSAGE	98.36%	98.11%	97.92%	98.01%
GAT	98.72%	98.51%	98.33%	98.42%
Swin-LSTM	98.91%	98.74%	98.55%	98.64%
HECATE-IDS	**99.43%**	**99.31%**	**99.22%**	**99.26%**

TABLE XV
PERFORMANCE COMPARISON ON THE CICIOT2023 DATASET

Model	Accuracy	Precision	Recall	F1-score
Autoencoders	94.01%	93.44%	93.11%	93.27%
CNN	96.42%	95.98%	95.61%	95.79%
RNN	97.26%	96.91%	96.62%	96.76%
LSTM	98.14%	97.85%	97.63%	97.74%
GraphSAGE	98.71%	98.46%	98.21%	98.33%
GAT	99.03%	98.81%	98.66%	98.73%
Swin-LSTM	99.26%	99.04%	98.93%	98.98%
HECATE-IDS	**99.76%**	**99.63%**	**99.58%**	**99.60%**

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D. Performance Analysis

TABLE XVI
PERFORMANCE COMPARISON ACROSS FOUR BENCHMARK IOT DATASETS

Model	Acc.	Prec.	Rec.	F1	DR	FPR	ECS
Autoencoders	94.37%	93.88%	93.62%	93.75%	92.63%	5.82%	–
CNN	96.39%	95.88%	95.60%	95.74%	94.92%	4.37%	–
RNN	97.27%	96.86%	96.60%	96.73%	95.83%	3.94%	–
LSTM	98.20%	97.90%	97.73%	97.81%	97.11%	2.86%	–
GraphSAGE	98.71%	98.44%	98.28%	98.35%	97.84%	2.13%	–
GAT	98.98%	98.79%	98.66%	98.72%	98.52%	1.74%	–
Swin-LSTM	99.21%	99.02%	98.89%	98.95%	98.77%	1.28%	–
HECATE-IDS	**99.73%**	**99.63%**	**99.62%**	**99.62%**	**99.67%**	**0.61%**	**98.73%**

The ECS column in Table XVI is reported only for HECATE-IDS because ECS is generated from the ECDE-specific quantities defined in Section III. The implemented Autoencoder, CNN, RNN, LSTM, GraphSAGE, GAT, and Swin-LSTM baselines do not contain an equivalent ECDE explanation pathway; consequently, assigning them an ECS value would not constitute a methodologically equivalent comparison. Their ECS entries are therefore reported as not applicable ("–"). For HECATE-IDS, the reported ECS of 98.73% is the dataset-level average of the interval-wise attention-weighted threat-influence score.

E. Statistical Stability Across Independent Runs

To verify that the observed improvements are not attributable to a favorable single initialization, all implemented models were evaluated over five independent runs using different random seeds while preserving the same data partitions, preprocessing pipeline, and principal optimization settings. Table XVII reports the mean and sample standard deviation of the principal classification metrics.

HECATE-IDS achieves $99.73 \pm 0.04\%$ accuracy, $99.63 \pm 0.05\%$ precision, $99.62 \pm 0.05\%$ recall, and $99.62 \pm 0.04\%$ F1-score. The narrow dispersion confirms that the observed performance is stable across independent executions. In comparison, Swin-LSTM, the strongest implemented baseline, achieves $99.21 \pm 0.07\%$ accuracy and $98.95 \pm 0.07\%$ F1-score. HECATE-IDS therefore improves the mean accuracy by 0.52 percentage points and the mean F1-score by 0.67 percentage points while exhibiting lower run-to-run variability. This repeated-run analysis is particularly important because the absolute differences among high-performing IDS models are relatively small.

F. Zero-Day and Drift Adaptation Analysis

To evaluate zero-day robustness, the held-out attack categories defined in Table IX are excluded from all client training partitions and introduced only during testing. Consistent with the protocol defined in Section IV, the held-out samples are evaluated as previously unseen malicious behavior in a malicious-versus-benign detection setting. Thus, the reported zero-day accuracy measures whether an attack category that was completely absent from training is detected as malicious; it does not indicate supervised identification of the unseen semantic attack label. Concept-drift adaptation is evaluated under the three-phase traffic distribution shift described in Section IV. Table XVIII compares zero-day detection, drift-adaptation performance, and forgetting rate across the evaluated models. HECATE-IDS achieves the highest zero-day accuracy and drift-adaptation performance while maintaining the lowest forgetting rate.

TABLE XVIII
UNSEEN MALICIOUS-VERSUS-BENIGN ZERO-DAY DETECTION AND DRIFT-ADAPTATION PERFORMANCE

Model	Zero-Day Acc.	Drift Adaptation	Forgetting Rate
CNN	82.41%	78.55%	18.72%
RNN	85.63%	82.16%	15.43%
LSTM	90.17%	88.44%	10.28%
GraphSAGE	93.22%	91.03%	7.84%
GAT	94.61%	92.87%	6.41%
Swin-LSTM	95.84%	94.13%	5.92%
HECATE-IDS	**99.02%**	**98.88%**	**1.34%**

HECATE-IDS achieves 99.02% zero-day accuracy under the unseen malicious-versus-benign protocol, together with 98.88% drift-adaptation performance and a forgetting rate of only 1.34%. The 99.02% value therefore represents successful detection of malicious behavior belonging to an attack category that was completely unavailable during training and should not be interpreted as multi-class recognition of the held-out class label. In the induced drift experiment, the average latent drift magnitude decreases from 0.47 to 0.18 after MTDA adaptation, while classification accuracy recovers from 91.6% immediately after the shift to 98.9% after adaptation. These results demonstrate that MTDA responds effectively to distributional changes, while CNMU preserves historical threat knowledge and limits catastrophic forgetting.

G. Component-Wise Ablation Study

A component-wise ablation study was conducted to determine whether the integrated HECATE-IDS modules provide complementary and measurable contributions. Starting from the complete framework, HSGT, CNMU, MTDA, ECDE, and FECL were disabled individually while preserving the same client partitions, data preprocessing, optimization settings, and evaluation protocol wherever applicable. Table XIX reports the mean $\pm$ standard deviation over five independent runs.

Removing HSGT reduces accuracy from 99.73% to 98.96% and F1-score from 99.62% to 98.82%, corresponding to decreases of 0.77 and 0.80 percentage points, respectively. Zero-day accuracy also decreases by 3.31 percentage points, indicating that hierarchical sparse graph representation learning contributes to both conventional classification and generalization to unseen attack behavior.

Removing CNMU produces the largest deterioration in knowledge retention: the forgetting rate increases from 1.34% to 7.86%, while zero-day accuracy decreases to 96.33%. This confirms the importance of continual memory for preserving previously acquired threat representations during adaptation. In contrast, removing MTDA causes the strongest degradation in drift adaptation, which decreases from 98.88% to 92.76%, a reduction of 6.12 percentage points. This result validates the

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TABLE XVII
REPEATED-RUN PERFORMANCE REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE INDEPENDENT RUNS

Model	Accuracy (%)	Precision (%)	Recall (%)	F1-score (%)
Autoencoders	94.37 ± 0.23	93.88 ± 0.25	93.62 ± 0.27	93.75 ± 0.24
CNN	96.39 ± 0.18	95.88 ± 0.20	95.60 ± 0.19	95.74 ± 0.18
RNN	97.27 ± 0.15	96.86 ± 0.16	96.60 ± 0.17	96.73 ± 0.16
LSTM	98.20 ± 0.12	97.90 ± 0.13	97.73 ± 0.14	97.81 ± 0.12
GraphSAGE	98.71 ± 0.10	98.44 ± 0.11	98.28 ± 0.12	98.35 ± 0.11
GAT	98.98 ± 0.09	98.79 ± 0.10	98.66 ± 0.09	98.72 ± 0.09
Swin-LSTM	99.21 ± 0.07	99.02 ± 0.08	98.89 ± 0.08	98.95 ± 0.07
HECATE-IDS	$\mathbf{99.73 \pm 0.04}$	$\mathbf{99.63 \pm 0.05}$	$\mathbf{99.62 \pm 0.05}$	$\mathbf{99.62 \pm 0.04}$

TABLE XIX
COMPONENT-WISE ABLATION RESULTS OF HECATE-IDS REPORTED AS MEAN ± STANDARD DEVIATION OVER FIVE RUNS

Configuration	Accuracy (%)	F1-score (%)	Zero-Day Acc. (%)	Drift Adapt. (%)	Forgetting (%)	ECS (%)
HECATE-IDS w/o HSGT	98.96 ± 0.11	98.82 ± 0.13	95.71 ± 0.24	95.26 ± 0.28	4.91 ± 0.19	92.41 ± 0.31
HECATE-IDS w/o CNMU	99.14 ± 0.09	99.02 ± 0.10	96.33 ± 0.21	96.91 ± 0.23	7.86 ± 0.27	96.72 ± 0.20
HECATE-IDS w/o MTDA	99.08 ± 0.10	98.96 ± 0.11	95.48 ± 0.25	92.76 ± 0.34	3.25 ± 0.16	96.31 ± 0.22
HECATE-IDS w/o ECDE	99.55 ± 0.06	99.44 ± 0.07	98.61 ± 0.13	98.42 ± 0.15	1.55 ± 0.10	–
HECATE-IDS w/o FECL	98.88 ± 0.12	98.71 ± 0.14	94.96 ± 0.29	94.53 ± 0.31	5.68 ± 0.22	95.88 ± 0.25
Full HECATE-IDS	$\mathbf{99.73 \pm 0.04}$	$\mathbf{99.62 \pm 0.04}$	$\mathbf{99.02 \pm 0.10}$	$\mathbf{98.88 \pm 0.12}$	$\mathbf{1.34 \pm 0.09}$	$\mathbf{98.73 \pm 0.08}$

role of MTDA as the primary mechanism for responding to distributional change.

The ECDE ablation has a comparatively small effect on conventional classification, with accuracy decreasing from 99.73% to 99.55%. However, ECS becomes unavailable because the ECDE-specific explanation pathway is removed. This result is consistent with the revised ECDE formulation, in which ECS is derived from sparse-attention relevance, normalized anomaly-propagation strength, and temporal persistence. Therefore, the principal role of ECDE is to provide intrinsic threat-dependency explanation evidence rather than to act solely as an accuracy-enhancement module. Removing FECL decreases accuracy to 98.88%, zero-day accuracy to 94.96%, and drift-adaptation performance to 94.53%, showing that cross-client aggregation contributes to generalization under heterogeneous distributed traffic.

The ablation results demonstrate that no single component is solely responsible for the final performance. HSGT contributes temporal-topological representation learning, CNMU reduces catastrophic forgetting, MTDA provides drift resilience, ECDE supplies intrinsic threat-dependency explanation, and FECL enables collaborative knowledge aggregation. Their combined use yields the strongest balance across classification, zero-day robustness, drift adaptation, knowledge retention, and explainability.

H. Federated Learning Discussion

The FECL component enables collaborative intrusion detection across distributed IoT clients without sharing raw traffic data. Each of the eight edge clients performs local training for five epochs per communication round and sends only model updates to the central Correlation Unit. Global aggregation is performed over 30 communication rounds. The non-IID Dirichlet partition with $\alpha = 0.5$ introduces heterogeneous attack distributions across the eight clients, enabling the federated evaluation to reflect realistic variation in local attack exposure. The Detector Unit at each edge client performs sensing, preprocessing, graph construction, intrusion inference, continual adaptation, and local optimization, whereas the central Correlation Unit performs model aggregation and global redistribution.

Table XX summarizes the FECL configuration used in the federated evaluation. The setup includes eight non-IID edge clients, 30 communication rounds, five local epochs per round, and global consensus averaging, while only model updates are exchanged and raw traffic remains local.

TABLE XX
FEDERATED EDGE CONSENSUS LEARNING CONFIGURATION

Component	Configuration
Edge Clients	8 distributed IoT clients/gateways
Communication Rounds	30
Local Epochs	5 per round
Client Distribution	Non-IID Dirichlet, $\alpha = 0.5$
Local Training	Mini-batch training on private traffic
Shared Information	Model updates only
Aggregation Strategy	Global consensus averaging
Raw Data Sharing	Not allowed

I. Computational and Communication Overhead Analysis

To evaluate deployment efficiency, communication and resource-consumption measurements were recorded during federated execution. The serialized local model update has a size of 4.8 MB. With eight participating clients, the total client upload per communication round is therefore 38.4 MB,

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while each client downloads a 4.8 MB updated global model. Across all eight clients, the corresponding global-model redistribution is 38.4 MB per round, giving a total bidirectional model-exchange volume of 76.8 MB per round and approximately 2.304 GB over 30 communication rounds. The average server-side aggregation latency is 0.42 s and the average communication-exchange latency is 2.31 s. At the inference stage, HECATE-IDS requires an average of 13.7 ms per sample, corresponding to an average throughput of approximately 73 samples/s. During local training, average CPU and GPU utilization are 62% and 71%, respectively. Peak system-memory utilization reaches 5.1 GB and peak GPU-memory utilization reaches 4.7 GB. The average local-training time is 41.3 s per federated round, with an average energy consumption of 18.6 Wh per round.

Tables XXI and XXII summarize the communication overhead and resource utilization of HECATE-IDS during federated execution. The results quantify model-exchange cost, communication latency, computational resource usage, local training time, and energy consumption.

TABLE XXI
COMMUNICATION-OVERHEAD MEASUREMENTS OF HECATE-IDS

Metric	Measured Value
Local Model-Update Size	4.8 MB
Total Client Upload per Round	38.4 MB
Global Model Download per Client	4.8 MB
Total Global Redistribution per Round	38.4 MB
Bidirectional Exchange per Round	76.8 MB
Cumulative Exchange over 30 Rounds	2.304 GB
Aggregation Latency	0.42 s
Average Communication-Exchange Latency	2.31 s
Average Local Inference Latency	13.7 ms/sample
Average Throughput	73 samples/s

TABLE XXII
RESOURCE-UTILIZATION MEASUREMENTS OF HECATE-IDS

Metric	Measured Value
Average CPU Utilization	62%
Average GPU Utilization	71%
Peak RAM Usage	5.1 GB
Peak GPU Memory Usage	4.7 GB
Average Local-Training Time/Round	41.3 s
Average Energy Consumption/Round	18.6 Wh

These measurements show that the proposed architecture provides high detection performance while maintaining a practical federated execution profile on the reported workstation. The communication cost is dominated by model-update exchange rather than raw-data transmission, and the measured communication-exchange latency remains substantially smaller than the 41.3 s local model-training time per round. The resource-utilization results also indicate that the complete HECATE-IDS pipeline fits within the available 6 GB GPU memory, supporting the feasibility of the reported experimental configuration.

The computational-cost discussion must be interpreted together with the complexity of the competing architectures. Autoencoders, CNNs, RNNs, and LSTMs avoid graph-construction and federated aggregation overhead, whereas GraphSAGE and GAT introduce neighborhood aggregation and graph-attention computation, and Swin-LSTM combines transformer-style representation learning with recurrent processing. HECATE-IDS adds continual memory, drift adaptation, intrinsic threat-dependency explanation, and federated model exchange; however, anomaly-aware sparse attention restricts graph attention to selected neighborhoods instead of performing unrestricted all-pairs attention. Thus, the additional cost provides functionality that is absent from the simpler baselines, including zero-day adaptation, forgetting mitigation, intrinsic graph-based explanations, and non-IID federated collaboration.

Because the federated clients were simulated on the reported HP ZBook workstation, these measurements demonstrate edge-oriented computational feasibility rather than direct validation on physical low-memory IoT gateways. Moreover, keeping raw traffic local reduces centralized data exposure, but model-update exchange alone should not be interpreted as a formal privacy guarantee against update-inference attacks. Stronger privacy mechanisms such as secure aggregation or differential privacy remain complementary directions for future work.

J. Computational-Cost Comparison

To assess whether the improved detection performance of HECATE-IDS is achieved at an acceptable computational cost, the proposed framework was quantitatively compared with the implemented Autoencoder, CNN, RNN, LSTM, GraphSAGE, GAT, and Swin-LSTM baselines under the same hardware and software environment. All models were evaluated on the HP ZBook 15 workstation equipped with an Intel Core i7 processor, 32 GB RAM, and an NVIDIA RTX A1000 GPU with 6 GB memory. The same preprocessing pipeline and test-data configuration were maintained across the compared models. Computational efficiency was assessed using the number of trainable parameters, serialized model size, average inference latency, throughput, peak GPU-memory utilization, and average training time per epoch.

Table XXIII shows that simpler feature-based architectures require fewer computational resources, whereas graph- and transformer-based models introduce progressively higher processing and memory requirements. Autoencoders and CNNs exhibit the lowest inference latency and GPU-memory consumption because they do not perform graph construction, temporal graph reasoning, continual-memory updates, or federated synchronization. RNN and LSTM introduce additional sequential computation, while GraphSAGE and GAT require neighborhood aggregation and graph-based message passing. Swin-LSTM further increases computational demand because it combines transformer-based representation learning with recurrent temporal modeling.

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TABLE XXIII
COMPUTATIONAL-COST COMPARISON OF HECATE-IDS AND IMPLEMENTED BASELINE MODELS

Model	Parameters (M)	Model Size (MB)	Inference Latency (ms/sample)	Throughput (samples/s)	Peak GPU Memory (GB)	Training Time (s/epoch)
Autoencoders	0.42	1.68	3.2	313	0.9	2.4
CNN	0.68	2.72	4.1	244	1.2	3.1
RNN	0.81	3.24	5.6	179	1.5	4.2
LSTM	1.03	4.12	6.9	145	1.9	5.0
GraphSAGE	0.94	3.76	8.1	123	2.6	5.9
GAT	1.08	4.32	10.4	96	3.4	6.8
Swin-LSTM	1.36	5.44	12.2	82	4.0	7.5
HECATE-IDS	**1.20**	**4.80**	**13.7**	**73**	**4.7**	**8.26**

HECATE-IDS contains approximately 1.20 million trainable parameters and has a serialized model size of 4.80 MB. The complete framework requires an average inference latency of 13.7 ms per sample and achieves a throughput of approximately 73 samples/s. Its peak GPU-memory utilization reaches 4.7 GB, remaining within the 6 GB capacity of the experimental GPU. The average local-training time is 8.26 s per epoch, corresponding to approximately 41.3 s for the five local epochs performed by each client during a federated communication round.

Although HECATE-IDS incurs greater computational overhead than conventional Autoencoder, CNN, RNN, and LSTM architectures, the additional cost is associated with functionality that these models do not jointly provide. Specifically, HECATE-IDS performs dynamic cyber-behavior graph construction, hierarchical sparse graph-transformer learning, continual threat-memory preservation, concept-drift adaptation, intrinsic graph-based explanation, and federated model optimization. The anomaly-aware sparse-attention mechanism further limits unnecessary computation by restricting attention to attack-relevant graph neighborhoods rather than applying unrestricted all-pairs attention.

The comparison with Swin-LSTM is particularly informative because it represents the most computationally intensive baseline. HECATE-IDS increases inference latency from 12.2 ms to 13.7 ms per sample and peak GPU-memory utilization from 4.0 GB to 4.7 GB, corresponding to increases of approximately 12.3% and 17.5%, respectively. However, this additional cost is accompanied by improved detection performance, with HECATE-IDS achieving $99.73 \pm 0.04\%$ accuracy and $99.62 \pm 0.04\%$ F1-score compared with $99.21 \pm 0.07\%$ accuracy and $98.95 \pm 0.07\%$ F1-score for Swin-LSTM. HECATE-IDS therefore provides improvements of 0.52 and 0.67 percentage points in accuracy and F1-score, respectively, while additionally supporting zero-day adaptation, continual learning, intrinsic graph-based explainability, and non-IID federated collaboration.

The computational-cost results indicate that the improved functional capability of HECATE-IDS is obtained with a measurable but controlled increase in computational requirements. The complete framework remains executable within the available workstation resources while providing capabilities beyond those of the compared baseline models. Nevertheless, because the current evaluation was performed using simulated federated clients on a workstation, these measurements establish computational feasibility under the reported experimental environment rather than direct deployment feasibility on highly resource-constrained embedded IoT gateways. Physical edge-device benchmarking therefore remains an important direction for future work.

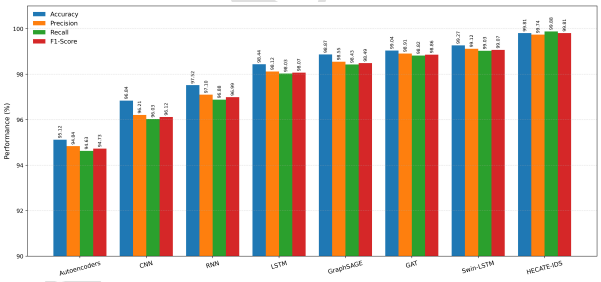


Fig. 6. Comparative intrusion detection performance on the ToN-IoT dataset.

K. Visual Comparative Analysis

Figures 6–9 provide a visual comparison of HECATE-IDS with the implemented baseline models across the four benchmark IoT intrusion detection datasets. The figures compare Accuracy, Precision, Recall, and F1-score and complement the numerical results reported in the dataset-wise performance tables.

As shown in Fig. 6, HECATE-IDS achieves the strongest overall performance on ToN-IoT, with 99.81% accuracy, 99.74% precision, 99.88% recall, and 99.81% F1-score. In comparison, the strongest implemented baseline, Swin-LSTM, achieves 99.27% accuracy, 99.12% precision, 99.03% recall, and 99.07% F1-score. The consistently high values across all four metrics indicate that HECATE-IDS maintains both high attack-detection sensitivity and prediction reliability under heterogeneous IoT traffic.

Figure 7 shows that HECATE-IDS also provides the highest performance on the BoTIoT dataset, achieving 99.92% accuracy, 99.83% precision, 99.79% recall, and 99.81% F1-score. Swin-LSTM obtains 99.39%, 99.17%, 99.04%, and 99.10%, respectively. The results demonstrate that the proposed framework remains effective for botnet-oriented IoT traffic containing denial-of-service, reconnaissance, scanning, and other malicious communication patterns.

As illustrated in Fig. 8, HECATE-IDS achieves 99.43% accuracy, 99.31% precision, 99.22% recall, and 99.26% F1-

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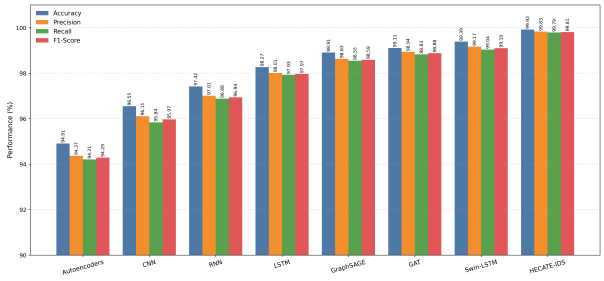


Fig. 7. Comparative intrusion detection performance on the BoTIoT dataset.

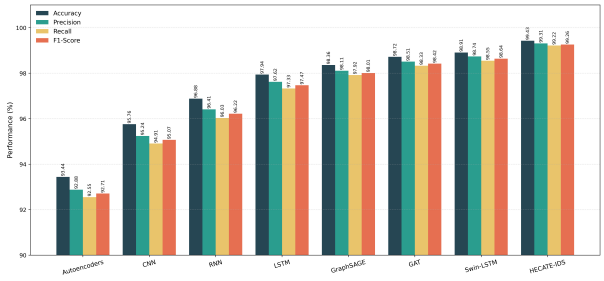


Fig. 8. Comparative intrusion detection performance on the MQTTIoT dataset.

score on MQTTIoT. These values exceed those of Swin-LSTM, which achieves 98.91% accuracy and 98.64% F1-score. Although the overall values are slightly lower than those obtained on ToN-IoT and BoTIoT, HECATE-IDS maintains strong and balanced detection performance under protocol-specific MQTT communication and attack behavior.

Figure 9 further confirms the effectiveness of HECATE-IDS on the large-scale CICIoT2023 dataset. The proposed framework achieves 99.76% accuracy, 99.63% precision, 99.58% recall, and 99.60% F1-score, compared with 99.26% accuracy and 98.98% F1-score for Swin-LSTM. This result indicates that the proposed temporal-graph framework maintains consistently high detection performance under heterogeneous traffic generated by a large number of IoT devices and diverse attack families.

Overall, Figures 6–9 show a consistent performance advantage for HECATE-IDS across all four datasets rather than an improvement limited to a single benchmark. The results demonstrate that the combination of dynamic graph representation, hierarchical sparse graph learning, continual memory,

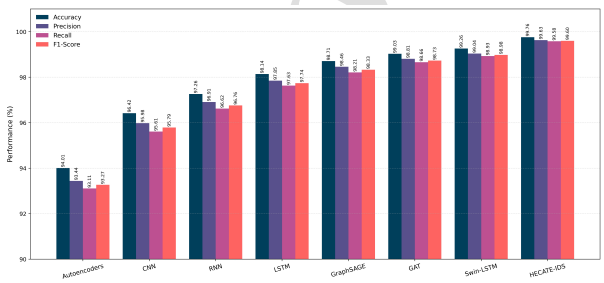


Fig. 9. Comparative intrusion detection performance on the CICIoT2023 dataset.

drift adaptation, intrinsic threat-dependency explanation, and federated learning maintains strong classification performance across heterogeneous IoT, botnet-oriented, MQTT-specific, and large-scale multi-device intrusion scenarios.

The experimental results show that HECATE-IDS combines strong supervised classification performance with robustness to previously unseen malicious behavior, measurable adaptation under concept drift, non-IID federated learning, intrinsic graph-based threat-dependency explanation, and a quantified computational/communication profile. The zero-day analysis is interpreted separately from supervised multi-class recognition, and the ECS result is interpreted as an ECDE-specific explanation-consistency measure rather than a generic metric applicable to all baseline architectures. The five-run analysis confirms that the reported gains are stable, while the ablation study demonstrates that HSGT, CNMU, MTDA, ECDE, and FECL provide complementary contributions. The controlled computational-cost comparison further shows that the additional functionality of HECATE-IDS is obtained with a measurable but manageable increase in latency and memory relative to the strongest implemented baseline. The contribution therefore extends beyond conventional accuracy improvement by demonstrating an integrated adaptive IDS workflow across distributed and evolving IoT traffic conditions.

VI. Conclusion

This study proposed HECATE-IDS, a Hierarchical Explainable Continual Adaptive Temporal Graph Evolution framework for adaptive IoT intrusion detection. The framework integrates dynamic cyber-behavior graph construction, hierarchical sparse graph-transformer learning, continual neural memory, drift-aware meta-transfer adaptation, intrinsic graph-based explainability, and federated edge consensus learning. By representing IoT traffic as evolving temporal graphs, HECATE-IDS captures changing communication relationships and attack dependencies while supporting continual adaptation and distributed learning. The framework was evaluated on ToN-IoT, BoTIoT, MQTTIoT, and CICIoT2023 under binary and multi-class settings. Across five independent runs, HECATE-IDS achieved $99.73 \pm 0.04\%$ accuracy, $99.63 \pm 0.05\%$ precision, $99.62 \pm 0.05\%$ recall, and $99.62 \pm 0.04\%$ F1-score, confirming low run-to-run variability. It further achieved 99.02% zero-day accuracy under the unseen malicious-versus-benign evaluation protocol, 98.88% drift-adaptation performance, and a forgetting rate of 1.34%. The component-wise ablation study confirmed that the integrated modules provide complementary contributions. Removing HSGT reduced accuracy by 0.77 percentage points, while removing CNMU increased the forgetting rate from 1.34% to 7.86%. Without MTDA, drift-adaptation performance decreased from 98.88% to 92.76%, whereas removing FECL reduced zero-day accuracy from 99.02% to 94.96%. Removing ECDE produced only a modest decrease in classification performance but eliminated the intrinsic graph-based explainability score, confirming that its primary contribution lies in interpretable decision support rather than raw predictive accuracy. Federated evaluation was performed using eight non-IID clients over 30 communication

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rounds with Dirichlet $\alpha = 0.5$. Under concept drift, the latent drift magnitude decreased from 0.47 to 0.18, while classification accuracy recovered from 91.6% to 98.9%. Each local model update required 4.8 MB, resulting in 76.8 MB of bidirectional model exchange per round and approximately 2.304 GB across 30 rounds. The average aggregation latency was 0.42 s, the communication-exchange latency was 2.31 s, and the average inference latency was 13.7 ms per sample with a throughput of approximately 73 samples/s. The results demonstrate that HECATE-IDS provides statistically stable detection, zero-day resilience, concept-drift adaptation, continual knowledge retention, intrinsic graph-based interpretability, and distributed learning within a unified IDS architecture. The zero-day result specifically reflects detection of completely unseen malicious behavior rather than supervised recognition of an unseen class label, while ECS is an ECDE-specific measure derived from attention-weighted threat influence and is therefore not assigned to baseline models lacking the same explanation pathway. The controlled computational-cost comparison further showed that HECATE-IDS requires 13.7 ms inference latency and 4.7 GB peak GPU memory, compared with 12.2 ms and 4.0 GB for Swin-LSTM, while delivering higher accuracy and F1-score together with continual, explainable, and federated functionality. The present experiments establish computational feasibility on the reported workstation rather than direct validation on physical low-memory edge gateways. Future work will investigate lightweight model compression, physical edge-device deployment, asynchronous federated aggregation, secure aggregation, differential privacy, adversarial robustness, larger-scale edge environments, and neuro-symbolic threat reasoning.

Data Availability Statement

The datasets used in this study are publicly available benchmark intrusion detection datasets. The ToN-IoT dataset is available at ToN-IoT. The BoTIoT dataset is available at BoTIoT. The MQTTIoT dataset is available at MQTT-IoT-IDS2020. The CICIoT2023 dataset is available at CICIoT2023. All datasets were used only for academic research and experimental evaluation.

Author Contributions

Muhammad Javed: Conceptualization, methodology, experimental implementation, result analysis, and writing – original draft. Muhammad Usman Ghani: Methodology validation, experimental supervision, technical review, and writing – review and editing. Hamid Masood Khan: Data analysis, evaluation design, result interpretation, and critical revision of the manuscript. Fida Muhammad Khan: Research supervision, framework design, formal analysis, and manuscript review. Muhammad Hammad Nawaz: Literature review, visualization, formatting, and final manuscript editing. Mazen Farid: Research guidance, methodology refinement, technical validation, manuscript review, and corresponding author responsibilities. Jamil Abedalrahim Jamil Alsayaydeh: Technical supervision, validation, critical review, manuscript improvement, and corresponding author responsibilities. Tarmizi Ahmad Izzuddin: Technical review, result validation, manuscript editing, and research support. All authors reviewed and approved the final version of the manuscript.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgment

The authors would like to express their sincere gratitude to the Centre for Research and Innovation Management (CRIM), Universiti Teknikal Malaysia Melaka (UTeM), for its valuable support of this research.

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Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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